

PatternBoost Three-Problem Updated Experiment Analysis

Complete 81-cell trajectories, exact certificate audit, and limitations study

Multi-level PatternBoost project

July 12, 2026

Evidence commits	20750a588a5b; 6f80dfec1a16; 87326f25dbec
Analysis dataset	docs/assets/replacement_81_final_20260712
Design	3 problems $\times$ 3 representations $\times$ 3 local searches $\times$ 3 surrogates
Nominal search budget	24 hours per cell
Repeated-seed axis	None

Abstract

This report gives the final, configuration-level analysis of the 81-cell PatternBoost experiment on three geometric extremal-search problems: the LP integrality gap for maximum independent set of rectangles (MISR), the LP integrality gap for stabbing unit squares by horizontal and vertical lines, and recursive guillotine hardness for disjoint rectangle packings. Every cell starts from freshly generated geometry, uses a nominal 24-hour search budget, has no repeated-seed axis, and receives no warm-started certificate. The best exact scores were 1.5000 for MISR, 1.5385 for unit-square stabbing, and 0.3333 for guillotine hardness. The post-run audit freshly re-evaluated the 45 replacement certificates and matched the 36 retained certificates by configuration, hash, and score to their prior exact audits: 81 of 81 rows passed and 0 failed. Beyond the endpoint scores, this report analyzes every strict improvement against search generation, elapsed time, and cumulative transformer-training epoch. The main conclusion is that the observed plateaus are primarily structural: representation constraints, discrete objective values, surrogate/exact-score misalignment, and limited compositional mechanisms dominate the effect of additional epochs. By 25% of each row’s available model epochs, the final configuration ranking already has descriptive Spearman correlation at least 0.899 in all three problems; median post-best plateaus occupy 64–98% of search generations, and learned-model samples account for only 0–1.5% of strict record improvements.

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1 Questions answered by this study

The experiment was designed to answer five distinct questions.

1. What exact value does each of the 81 component combinations reach under the same nominal 86,400-second wall-clock budget?
2. How does each row improve as search generations and transformer-training epochs accumulate?
3. Which representations, local-search operators, and surrogates are associated with stronger final values, faster discovery, and longer plateaus?
4. Does the learned model contribute new record candidates, or does most progress come from initialization and local mutation?
5. Why did the search fail to move beyond the simple observed barriers, and which limitations are computational rather than mathematical?

The report deliberately separates three kinds of statements. An **exact result** is recomputed from a standalone certificate. A **descriptive matrix comparison** compares the 27 different configurations for one problem. A **mechanistic interpretation** uses both the trajectories and the implementation to explain a pattern. Only the first category is a certificate-level mathematical claim.

2 Experimental design and evidence standard

2.1 The full factorial matrix

For each problem, the matrix crosses three representations, three local-search operators, and three surrogate objectives. Thus each problem has $3^3 = 27$ cells and the complete experiment has 81 cells. This is a full component factorial, not 27 repetitions of one algorithm.

Provenance remark. The final table combines 36 retained cells from the July 9 matrix, 42 replacement cells from the main July 11 update, and three resolution-island unit-square cells from its final follow-up. All cells have the same nominal 24-hour wall-clock budget, but the replacement regime changes object count, population size, and unit-square diversity controls. The combined table is therefore the final algorithm comparison, not a pure one-factor causal ablation between the retained and replacement components.

The retained rows use $n = 8$, grid 8, search range 8–24, population 32, elite size 6, and model training every seven generations. In the 42-row replacement cohort, MISR and guillotine use $n = 12$, grid 8, and range 12–36; unit-square uses $n = 20$, grid 16, and range 20–60; all use population 16, elite size 4, and training every ten generations. The final three unit-square rows keep the same size/grid regime but use population 32, elite size 12, an initial pool of 128, four immigrants per generation, and resolution-island retention. Every regime uses exact scoring every five generations, 16 model samples, three model epochs per call, and checkpointing every generation.

2.2 What every component means

Across the complete study there are nine representation families, nine problem-specific local-search slots (seven distinct operator names), and nine surrogate objectives. Each problem uses three of each; only

Table 1: Component registry for the final matrix.

Problem	Representations	Local search	Surrogates
MISR	<code>triangle_free_rect</code> ; <code>quadratic_program_rectangles</code> ; <code>fixed_symmetry_rectangles</code>	<code>sequence_pair_pivot</code> ; <code>program_coeff_pivot</code> ; <code>symmetry_crossover_hillclimb</code>	<code>exact_lp_gap_pressure</code> ; <code>triangle_free_exact_gap_pressure</code> ; <code>graph_conflict_proxy</code>
Unit-square	<code>line_square_incidence</code> ; <code>sqstab_exact_grid</code> ; <code>fixed_symmetry_grid</code>	<code>coord_mutation</code> ; <code>sqstab_local_hillclimb</code> ; <code>symmetry_crossover_hillclimb</code>	<code>greedy_partial_lp_bitset</code> ; <code>exact_stab_gap_pressure</code> ; <code>incidence_statistics</code>
Guillotine	<code>rect_direct_disjoint</code> ; <code>recursive_obstruction_grammar</code> ; <code>fixed_symmetry_packing</code>	<code>packing_resize</code> ; <code>witness_breaking</code> ; <code>symmetry_crossover_hillclimb</code>	<code>first_cut_obstruction</code> ; <code>depth_limited_dp</code> ; <code>k_subset_nonseparability</code>

those three are crossed, giving $3 \times 3 \times 3 = 27$ valid cells per problem. A representation specifies *how an instance is encoded and repaired*; local search specifies *how an encoded instance is changed*; and a surrogate specifies *how candidates are ranked before expensive exact verification*. The exact verifier, not a surrogate, decides whether a certificate becomes a saved record.

Table 2: Representation glossary.

Problem	Representation	Meaning
MISR	<code>triangle_free_rect</code>	Direct rectangle coordinates, with generation and repair restricted so that the rectangle intersection graph is triangle-free. It searches sparse conflict graphs directly.
MISR	<code>quadratic_program_rectangles</code>	A bounded integer coefficient program, decoder choice, and block composition are decoded to rectangles and then repaired to the permitted size and triangle-free constraint.
MISR	<code>fixed_symmetry_rectangles</code>	Random direct rectangle coordinates at a fixed object count, canonicalized under translation, horizontal and vertical reflection, and axis exchange. Repair preserves the cardinality and valid MISR geometry.
Unit-square	<code>line_square_incidence</code>	A balanced row/column incidence layout is generated first and then mapped to square coordinates. The stored payload tracks which abstract grid row and column each square occupies.
Unit-square	<code>sqstab_exact_grid</code>	Distinct finite-grid square coordinates plus a common side length, normalized to a canonical origin after repair. It is designed to make the exact finite line-stabbing geometry easy to audit.
Unit-square	<code>fixed_symmetry_grid</code>	Distinct grid positions at fixed common side length $Q = 2$ and fixed object count, canonicalized under the same objective-preserving planar symmetries. Its nine final cells use 20 squares on a grid of extent 16.
Guillotine	<code>rect_direct_disjoint</code>	Direct coordinates for disjoint rectangles, followed by geometric repair that removes overlaps and enforces the size bounds.
Guillotine	<code>recursive_obstruction_grammar</code>	Randomly instantiated non-sliceable obstruction cores are combined with filler rectangles and then repaired. It is a random generator family, not a fixed injected certificate.

Problem	Representation	Meaning
Guillotine	<code>fixed_symmetry_packing</code>	A fixed-cardinality random disjoint-rectangle packing, canonicalized under translation, reflection, and axis exchange. It supplies no fixed obstruction or target score.

Table 3: Local-search glossary.

Problem	Local search	Meaning
MISR	<code>sequence_pair_pivot</code>	Uses representation-safe add, replace, nudge, resize, and delete proposals, with add/replace moves upweighted in direct triangle-free coordinates. The historical name remains, but no final representation stores endpoint sequences.
MISR	<code>program_coeff_pivot</code>	When a program payload is present, perturbs coefficient blocks, decoder choice, composition, or active size. In other representations it falls back to their safe local mutation templates.
MISR	<code>symmetry_crossover_hillclimb</code>	Runs 25 exactly scored, proposals using one-step shifts, random relocations, coordinate exchange between two rectangles, and objective-preserving planar symmetries. A lexicographic key prioritizes admissibility and then exact objective score; for program encodings the analogous moves act on coefficient blocks.
Unit-square	<code>coord_mutation</code>	Basic coordinate-level move, duplicate, or deletion of a square, followed by canonical geometric repair.
Unit-square	<code>sqstab_local_hillclimb</code>	Builds several nearby line-tightening, addition/replacement, deletion, restart, and canonical-shift candidates, then retains the strongest quick local neighbor.
Unit-square	<code>symmetry_crossover_hillclimb</code>	Applies the same 25-proposal exact hill climb: 55% one-step shifts, 25% random relocations, 10% pairwise coordinate exchange, and 10% random planar symmetry. Repair keeps all squares congruent; fixed cardinality is guaranteed only when paired with the fixed-symmetry representation.
Guillotine	<code>packing_resize</code>	Moves, resizes, adds, replaces, and occasionally bars rectangles, then applies a nonseparability-guided local step. It is a geometry-first packing search.
Guillotine	<code>witness_breaking</code>	Uses a separable-subset/cut witness to close a gap or add a bridge/blocker that makes the witness’s easy guillotine cut less effective.
Guillotine	<code>symmetry_crossover_hillclimb</code>	Uses the shared exact nonworsening hill climb on disjoint rectangles, with shifts, relocations, pairwise interval exchange, and planar symmetries followed by packing repair.

Table 4: Surrogate-objective glossary.

Problem	Surrogate	Meaning
MISR	<code>exact_lp_gap_pressure</code>	Calls the MISR exact scorer during ranking and uses the LP/integral gap, with a small triangle-free lower-bound tie-break when applicable. It is the most aligned, most expensive MISR ranking signal.
MISR	<code>triangle_free_exact_gap_pressure</code>	Rejects non-triangle-free candidates, then ranks the remaining candidates by their exact MISR gap with small density and certified-lower-bound tie-breaks.
MISR	<code>graph_conflict_proxy</code>	A cheap graph statistic based on intersection-edge density, a maximum-independent-set hint, maximum clique size, and a triangle-free bonus.
Unit-square	<code>greedy_partial_lp_bitset</code>	Approximates the gap by dividing a greedy stabbing cover size by a max-line-frequency fractional lower bound computed from incidence bitsets.
Unit-square	<code>exact_stab_gap_pressure</code>	Calls the exact stabbing solver during ranking and combines the exact gap with small terms favoring a large integral cover, low LP value, and nontrivial LP support.
Unit-square	<code>incidence_statistics</code>	Uses greedy cover size, the number of candidate stabbing lines, and balance of line frequencies; it is a cheap incidence-only proxy.
Guillotine	<code>first_cut_obstruction</code>	Measures how many rectangles the best first axis-parallel cut must cross, with bonuses for balanced partitions and comparable horizontal/vertical projection spans.
Guillotine	<code>depth_limited_dp</code>	Runs a depth-four dynamic-programming estimate of how many rectangles a shallow guillotine strategy can save, and ranks by its implied destroyed fraction.
Guillotine	<code>k_subset_nonseparability</code>	Tests whether majority-sized subsets resist guillotine separation (exactly for small n , sampled otherwise) and combines that pressure with shallow-DP, beam, balance, projection, and size terms.

Each cell began from independently generated pseudorandom geometry. The RNG seed and state recorded in the source checkpoints support provenance and replay, but there is no seed-replication axis and no user-selected starting seed: there is one run per component cell. Consequently, marginal means and rank correlations in this report are descriptive summaries of the matrix. They are not confidence intervals over repeated stochastic trials.

2.3 PatternBoost loop

Every cell repeatedly performs the following cycle.

1. Generate or repair a population in the cell’s representation.
2. Score every novel candidate with the chosen surrogate.
3. Exactly audit the elite candidates at the configured interval.
4. Save a certificate only when the exact score strictly improves.

5. Mutate elites with the chosen local operator.
6. Every seven retained-row generations or ten replacement-row generations, initialize and train a new small transformer for three epochs on the strongest valid archive entries and sample additional candidates from it.

Exact evaluation is not limited to the periodic elite audit. Exact-pressure surrogates invoke the exact scorer during ranking, and each `symmetry_crossover_hillclimb` mutation exact-scores 25 internal proposals.

The final three `sqstab_exact_grid/symmetry_crossover_hillclimb` unit-square cells add representation-specific diversity controls around this loop: a 128-candidate initial pool, four fresh immigrants per generation, and retention stratified by every available integer side length. No target score, certificate, preferred denominator, or preferred side length is supplied. The purpose is to keep multiple coordinate resolutions alive long enough for local search to test them.

The model never sees a target answer. Its training corpus is the serialized archive selected by the current surrogate. Exact verification remains outside the model and controls certificate acceptance. Model parameters are not carried from one training event to the next; each event is a fresh three-epoch fit. When transformer sampling does not yield enough parseable candidates, the implementation may fill the shortfall with an archive-trained character n-gram sampler.

2.4 What “epoch” means in the plots

Three progress axes are retained because they answer different questions.

- **Search generation** counts complete population updates. It is the most direct algorithmic progress variable, but expensive scorers complete fewer generations per hour.
- **Elapsed time** is the fairest resource comparison because every cell received the same wall-clock budget.
- **Cumulative transformer-training epoch** counts three epochs for every successful model-training event. Because each event trains a newly initialized model, this is an accounting axis over repeated short fits, not a standard learning curve for one continuously trained network. It should not be confused with search generation. A score can improve between training events through local mutation.

All three views are generated. The manuscript displays the model-epoch and generation panels for each problem, while the elapsed-time panels remain in the committed analysis directory. The model-epoch plot measures where improvements occur relative to learned-model updates; the generation and time views guard against misleading conclusions caused by different scorer speeds.

2.5 Nontriviality and natural-search constraints

The final code enforced at least eight geometric objects in every accepted training example. Unit-square certificates additionally required integer stabbing number at least four, and guillotine certificates required at least two destroyed rectangles. No row started from a previously saved best certificate. Seventy-seven rows saved a best certificate through ordinary initialization or search. Four replacement guillotine rows using the new `symmetry/crossover` local search reached finalization without such a

record and therefore exported a fresh structural-random fallback at score 1/6. These four rows remain in the matrix as poor outcomes; they are not used to raise a reported record.

The search is not entirely prior-free. Fixed-cardinality symmetry canonicalization, balanced incidence layouts, triangle-free repair, and recursive obstruction layouts encode domain knowledge. In particular, the guillotine `recursive_obstruction_grammar` contains a randomized geometric realization of a fixed seven-rectangle spiral topology. It does not encode a score or final certificate, but it is a strong construction-family prior and must be disclosed when interpreting representation contrasts.

3 Run integrity and exact audit

The deployed evidence comes from commits `20750a588a5b`, `6f80dfec1a16`, and `87326f25dbec`. The final compact dataset retains the matrix identifiers, event-derived trajectories, fields derived from source checkpoints and summaries, certificate metadata, and source-cohort provenance needed to reconstruct the 81 unique cells. The strict post-processing checked:

1. exactly 81 unique component cells and 27 cells per problem;
2. one completed summary and checkpoint per cell;
3. clean stop reason, return code, and elapsed-time fields;
4. a nonempty best-certificate path for every cell;
5. agreement between event-stream, checkpoint, summary, and certificate scores;
6. certificate hash consistency; and
7. independent exact verification of every best certificate.

All 81 selected tasks completed with return code zero and stop reason `budget_exhausted`. Observed row elapsed times range from 86,403 to 86,509 seconds around the nominal 86,400-second budget. Every row has a summary, checkpoint, terminal score, and recoverable improvement trajectory.

The audit outcome was 81 passed and 0 failed out of 81 rows. The 45 replacement certificates were freshly recomputed; the 36 retained rows were matched by configuration, hash, and score to their prior passed exact audits. A row that fails any of these checks is retained in the diagnostic appendix but is not evidence for a reported construction.

4 Overall results

The best values alone do not describe the experiment. MISR has median 1.3000 and range 1.0000–1.5000; unit-square stabbing has median 1.4141 and range 1.2500–1.5385; guillotine hardness has median 0.2500 and range 0.1667–0.3333. The number of cells attaining the best value is 1, 1, and 1, respectively. These differences distinguish a robust plateau from a rare configuration-specific hit.

Table 5: Overall outcome of the completed 81-cell matrix. Quantiles are across the 27 component configurations, not repeated random seeds.

Problem	Min.	Q1	Median	Q3	Best	Cells at best
MISR	1	$\frac{49}{40}$	$\frac{13}{10}$	$\frac{4}{3}$	$\frac{3}{2}$	1
Unit-square	$\frac{5}{4}$	$\frac{4}{3}$	1.4141	$\frac{3}{2}$	$\frac{20}{13}$	1
Guillotine	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{3}$	1

Final exact score for every matrix cell

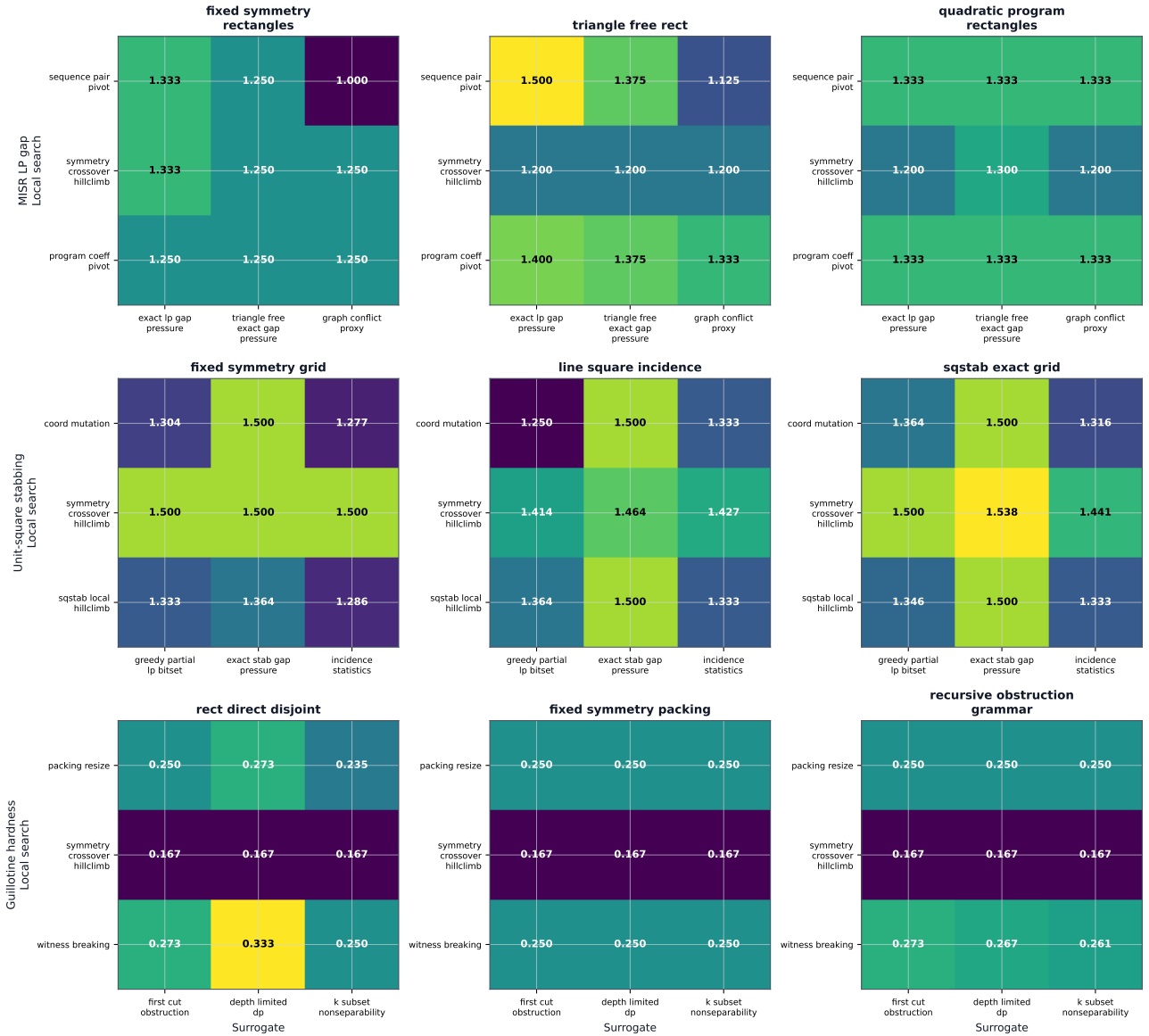


Figure 1: Final exact score in every cell. Rows within each panel are local searches, columns are surrogates, and the three panels in each problem row are representations. Color ranges are normalized within a problem; printed numbers are the evidence.

Table 6: Top five configurations per problem. Epoch is cumulative transformer training epoch; time is elapsed hours to the first appearance of the final best.

Problem	Score	Representation	Local search	Surrogate	Epoch	Time	Items
MISR	$\frac{3}{2}$	triangle_free_rect	sequence_pair_pivot	exact_lp_gap_pressure	1860	3.12	12
MISR	$\frac{7}{5}$	triangle_free_rect	program_coeff_pivot	exact_lp_gap_pressure	4782	14.48	14
MISR	$\frac{11}{8}$	triangle_free_rect	program_coeff_pivot	triangle_free_exact_gap_pressure	156	0.15	11
MISR	$\frac{11}{8}$	triangle_free_rect	sequence_pair_pivot	triangle_free_exact_gap_pressure	231	0.22	11
MISR	$\frac{4}{3}$	fixed_symmetry_rectangles	symmetry_crossover_hillclimb	exact_lp_gap_pressure	12	0.30	12
Unit-square	$\frac{20}{13}$	sqstab_exact_grid	symmetry_crossover_hillclimb	exact_stab_gap_pressure	54	0.37	20
Unit-square	$\frac{3}{2}$	sqstab_exact_grid	coord_mutation	exact_stab_gap_pressure	117	0.35	14
Unit-square	$\frac{3}{2}$	fixed_symmetry_grid	symmetry_crossover_hillclimb	incidence_statistics	183	0.92	20
Unit-square	$\frac{3}{2}$	sqstab_exact_grid	symmetry_crossover_hillclimb	greedy_partial_lp_bitset	39	1.41	20
Unit-square	$\frac{3}{2}$	fixed_symmetry_grid	symmetry_crossover_hillclimb	exact_stab_gap_pressure	408	2.06	20
Guillotine	$\frac{1}{3}$	rect_direct_disjoint	witness_breaking	depth_limited_dp	858	5.81	9
Guillotine	$\frac{3}{11}$	rect_direct_disjoint	packing_resize	depth_limited_dp	345	1.92	11
Guillotine	$\frac{3}{11}$	recursive_obstruction_grammar	witness_breaking	first_cut_obstruction	468	2.54	11
Guillotine	$\frac{3}{11}$	rect_direct_disjoint	witness_breaking	first_cut_obstruction	696	3.06	11
Guillotine	$\frac{4}{15}$	recursive_obstruction_grammar	witness_breaking	depth_limited_dp	669	8.10	15

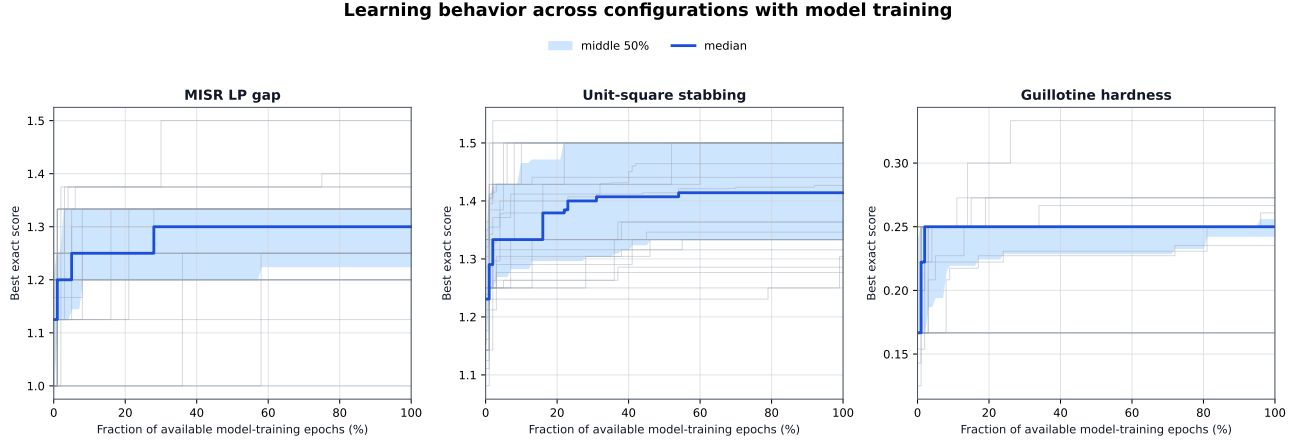


Figure 2: Trajectories after normalizing each trained cell by its available cumulative transformer-training epochs. Gray lines are individual configurations; the blue line is the median and the band is the middle 50%. The four zero-epoch guillotine fallback rows are omitted. This normalization compares the shape of learning, not raw computational throughput.

5 Learning dynamics across all 81 cells

5.1 Normalized model-epoch behavior

A long horizontal segment means that additional model updates and local search failed to improve the exact record. A late jump can mean either that the model found a new structural combination or that local mutation crossed a rare discrete objective threshold. The source analysis below separates those cases.

5.2 Early prediction of the final ranking

At 25% of each row’s available model epochs, the descriptive rank correlations with the final ranking are 0.908 for MISR, 0.899 for unit-square stabbing, and 0.944 for guillotine hardness. High early correlation indicates that longer runtime mostly refines an ordering already visible early; low correlation indicates meaningful late rearrangement among configurations.

All three correlations exceed 0.899 by this point. Unit-square and guillotine rankings are already strongly predictive after only 10% of each row’s available epochs, while MISR reorganizes somewhat longer. The remaining runtime mostly extends plateaus or improves a small number of outlying cells rather than changing the broad component ordering.

5.3 Where strict improvements came from

The learned-model path contributes to discovery only when it generates a valid candidate that survives representation repair, satisfies nontriviality, and beats the exact incumbent. The event schema labels these candidates `model_sample`; it does not distinguish a direct transformer sample from an n-gram backfill. The corresponding fractions of all strict improvements are 0.0%, 0.5%, and 1.5%. This statistic is more informative than the raw number of training calls: a model can train thousands of times while mostly reproducing the archive or proposing invalid candidates. In absolute terms, the event streams contain no model-sourced strict MISR improvement and one each for unit-square stabbing and guillotine.

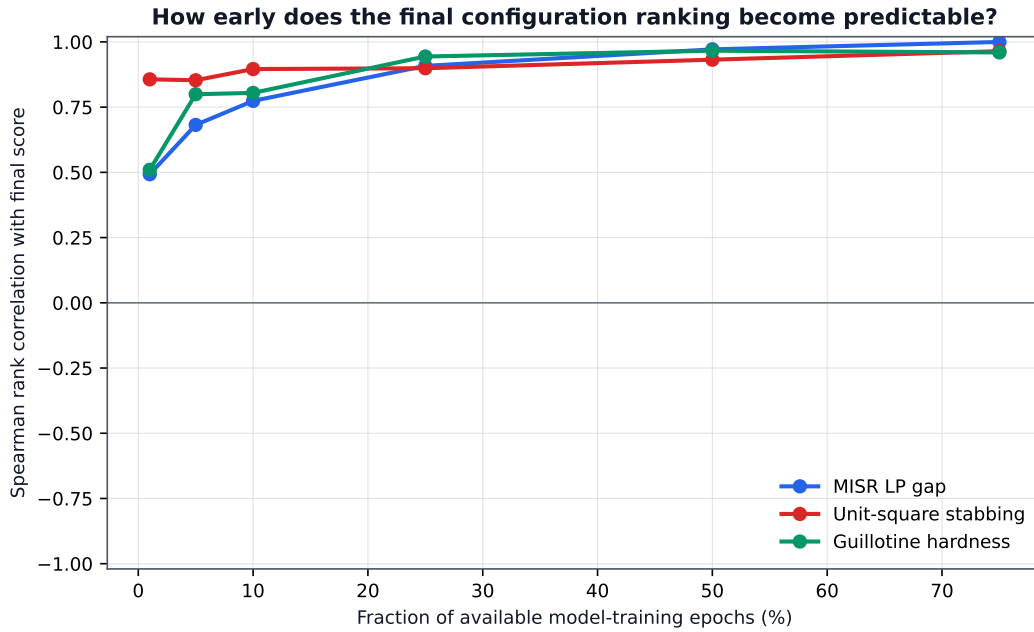


Figure 3: Spearman rank correlation between the configuration ranking at an early fraction of each row’s available model epochs and the final ranking. Rows with zero model-training epochs are undefined on this axis and omitted. Because each cell is a different configuration and there are no repeated seeds, the curve is descriptive rather than an inferential test.

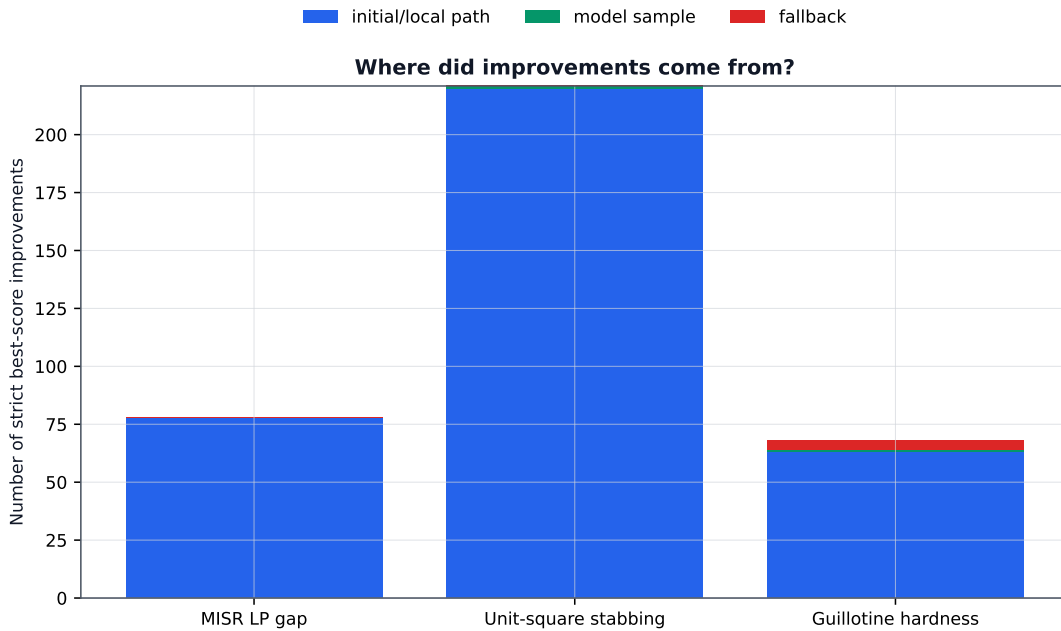


Figure 4: Source of every strict best-score improvement recorded in the 81 event streams. A model sample is counted only when that sampled candidate is exactly audited and strictly improves the saved record.

The combined initialization/local-mutation path accounts for 78, 220, and 63 strict improvements, respectively; four additional guillotine records are structural-random fallback exports. The combined label is deliberate: `initial_or_local` events do not support a finer attribution.

5.4 Plateaus and computational throughput

The median post-best plateau occupies 97.5% of MISR generations, 64.1% of unit-square generations, and 87.2% of guillotine generations. The top records first appeared after 3.12, 0.37, and 5.81 hours. The all-row guillotine median includes four terminal fallback exports assigned zero plateau; among the 23 ordinary-record rows its median plateau is 97.1%.

The median fractions of candidate slots rejected as duplicates were 42.6% for MISR, 36.1% for unit-square stabbing, and 22.6% for guillotine. Slot fractions are used because populations are 16 or 32 according to runtime regime. Median valid learned-model yields were only 11.1% and 26.0% for the two LP-gap problems, versus 86.7% for guillotine. High syntactic validity therefore does not imply useful novelty: guillotine proposals were often repairable, yet their exact scores still plateaued.

Plateau diagnostics

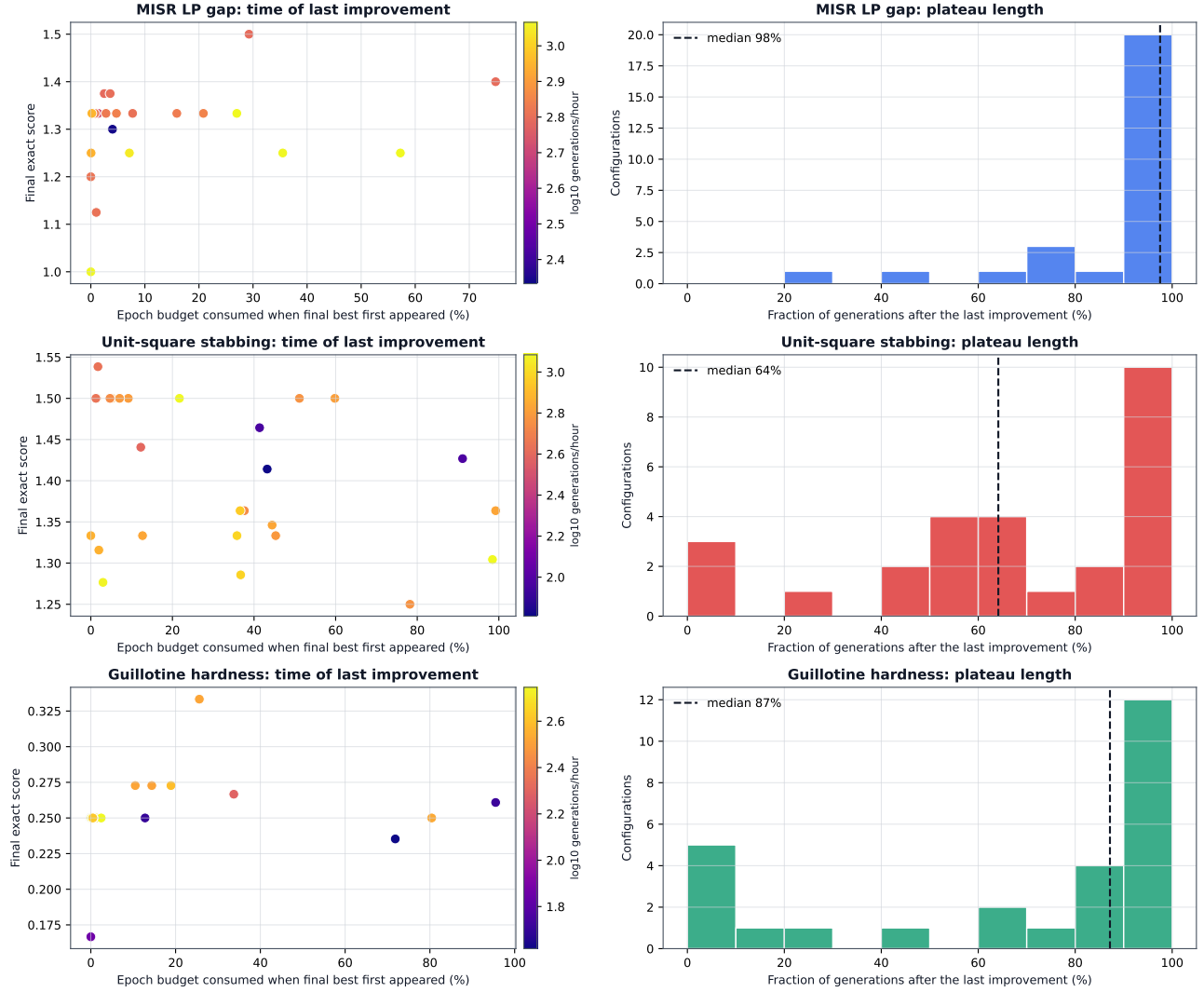


Figure 5: Left: when each cell first reached its final score, colored by search throughput. Right: fraction of search generations after the final strict improvement. Long plateaus show stasis under the observed search process; they do not by themselves establish an extended-run counterfactual.

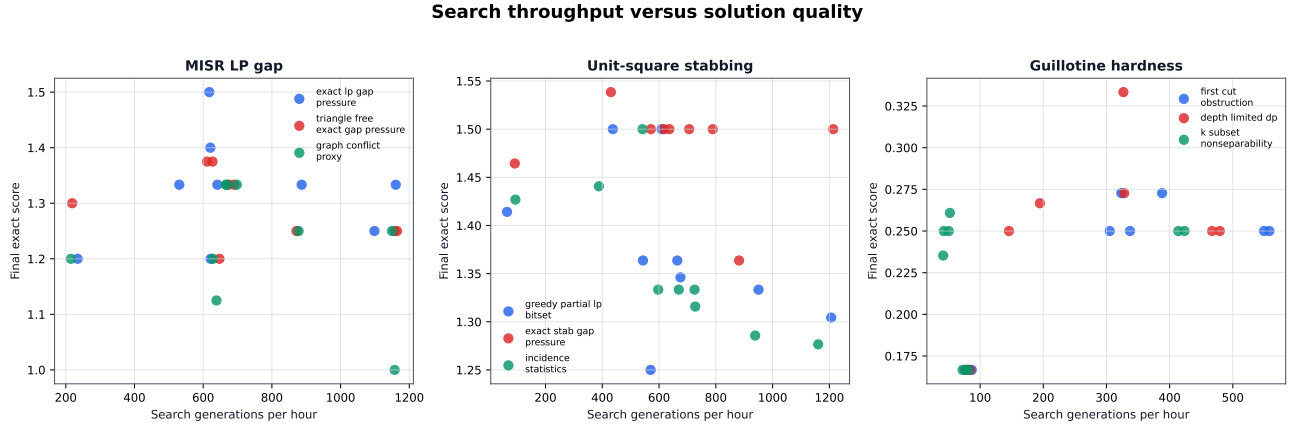


Figure 6: Search generations completed per hour versus final exact score. Exact or deeper surrogates may lower throughput while improving alignment; fast proxies are useful only if their additional generations translate into better exact certificates. Population, object count, training cadence, and internal exact-scoring cost also differ by runtime regime, so this is a descriptive diagnostic rather than a controlled speed comparison.

Search-efficiency diagnostics: duplication, model yield, and repair pressure

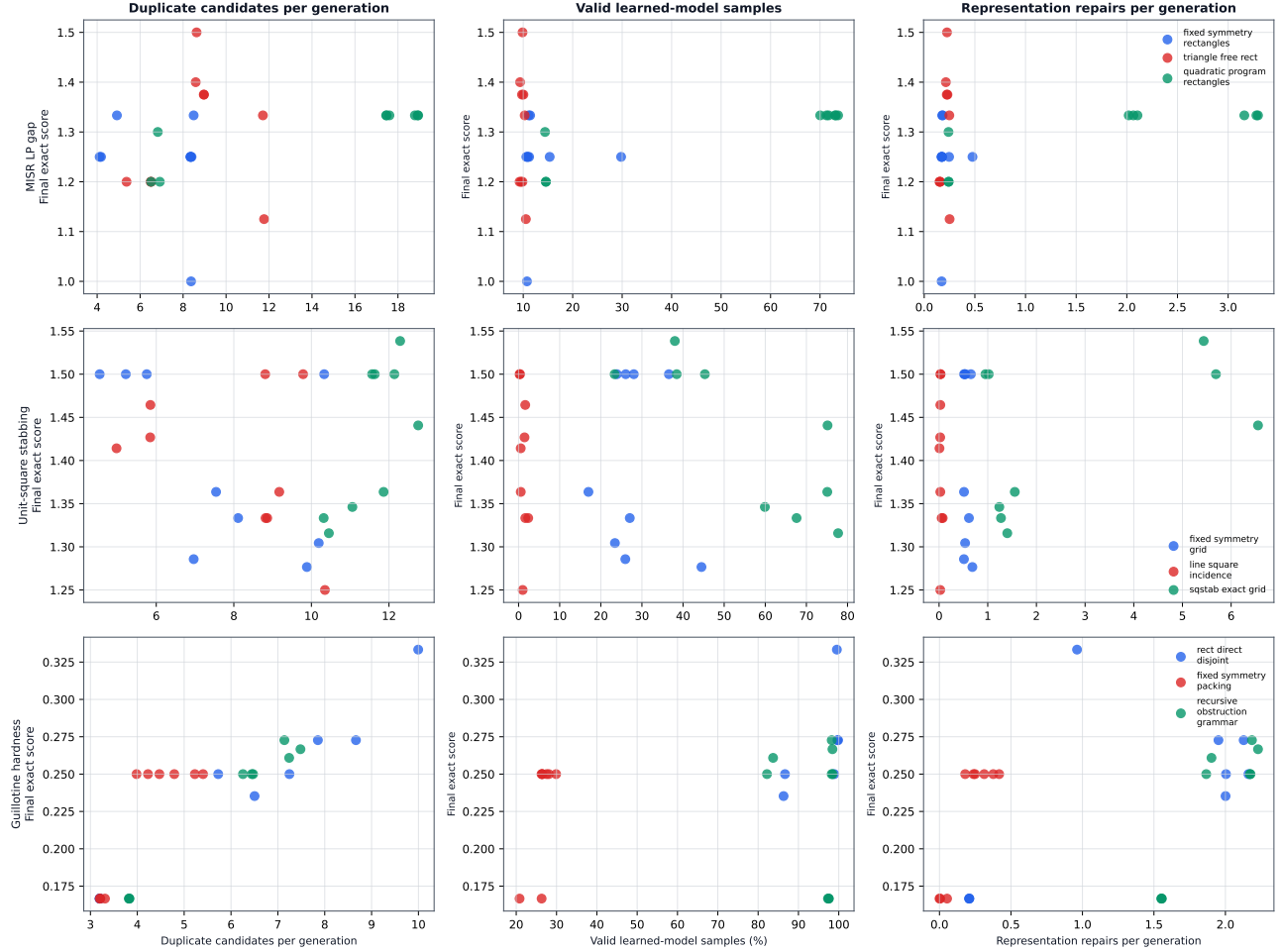


Figure 7: Final score against duplicate pressure, valid learned-model sample yield, and representation-repair pressure. These diagnostics test whether additional training produces genuinely new feasible structures or repeatedly returns to the same basin. Runtime regime and population size are potential confounders across component levels.

Table 7: Descriptive component marginals for MISR. Each level pools nine distinct component cells.

Factor	Level	Mean	Median	Min.	Max.	IQR
Representation	<code>triangle_free_rect</code>	1.3009	1.3333	1.1250	1.5000	[1.2000, 1.3750]
	<code>quadratic_program_rectangles</code>	1.3000	1.3333	1.2000	1.3333	[1.3000, 1.3333]
	<code>fixed_symmetry_rectangles</code>	1.2407	1.2500	1.0000	1.3333	[1.2500, 1.2500]
Local Search	<code>program_coeff_pivot</code>	1.3176	1.3333	1.2500	1.4000	[1.2500, 1.3333]
	<code>sequence_pair_pivot</code>	1.2870	1.3333	1.0000	1.5000	[1.2500, 1.3333]
	<code>symmetry_crossover_hillclimb</code>	1.2370	1.2000	1.2000	1.3333	[1.2000, 1.2500]
Surrogate	<code>exact_lp_gap_pressure</code>	1.3204	1.3333	1.2000	1.5000	[1.2500, 1.3333]
	<code>triangle_free_exact_gap_pressure</code>	1.2963	1.3000	1.2000	1.3750	[1.2500, 1.3333]
	<code>graph_conflict_proxy</code>	1.2250	1.2500	1.0000	1.3333	[1.2000, 1.3333]

6 Problem 1: MISR LP-gap search

6.1 Objective and exact verifier

For a finite family $\mathcal{R}$ of axis-parallel rectangles, the exact score is

$$\text{score}_{\text{MISR}}(\mathcal{R}) = \frac{\alpha^*(\mathcal{R})}{\alpha(\mathcal{R})},$$

where α is the maximum number of pairwise-disjoint rectangles and α^* is the clique-LP optimum of the rectangle intersection graph. The certificate stores all rectangles, the intersection graph, maximal cliques, an optimal integral independent set, LP primal and dual values, and solver status. The verifier reconstructs and solves both optimization problems.

6.2 All 27 trajectories

The matrix shows descriptive marginal contrasts, not clean causal effects. The `triangle_free_rect` representation has mean score 1.3009 and is the only representation to reach $3/2$. The `quadratic_program_rectangles` representation is nearly as strong on mean (1.3000) but remains between $6/5$ and $4/3$, while the new `fixed_symmetry_rectangles` representation averages 1.2407. Surrogate alignment is visible: `exact_lp_gap_pressure` averages 1.3204, compared with 1.2250 for `graph_conflict_proxy`. The local search axis also matters in the updated table: `program_coeff_pivot` averages 1.3176, `sequence_pair_pivot` 1.2870, and the new `symmetry_crossover_hillclimb` 1.2370.

6.3 What worked

The strongest MISR behavior combines a representation that can maintain sparse conflict structure with a surrogate that observes the exact LP/integer imbalance. Triangle-free repair removes three-way cliques, making edge constraints the clique LP and exposing half-integral pressure. In the winning direct

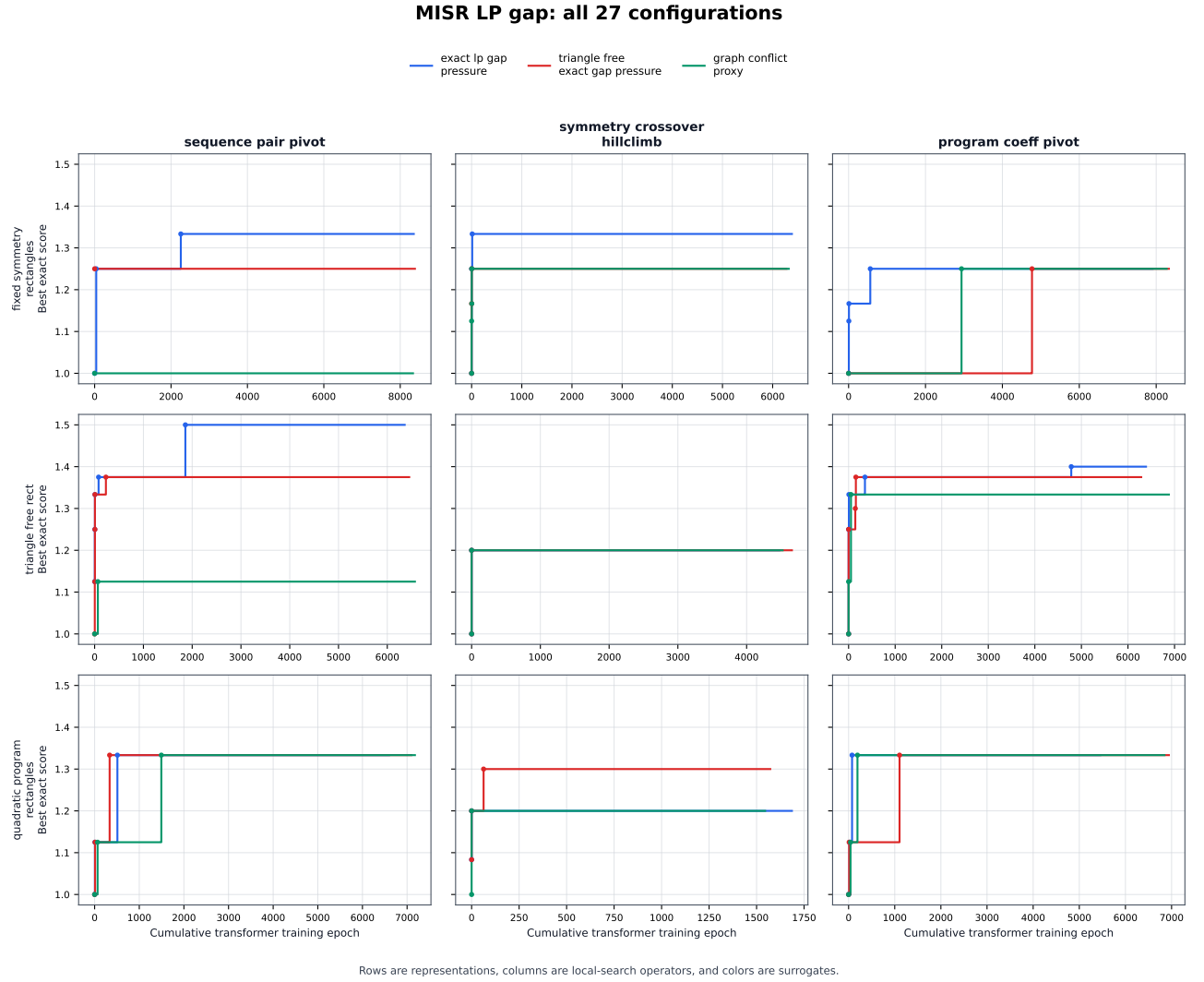


Figure 8: MISR trajectories against cumulative transformer-training epoch. Each panel fixes a representation and local search; its three colored curves are the surrogates.

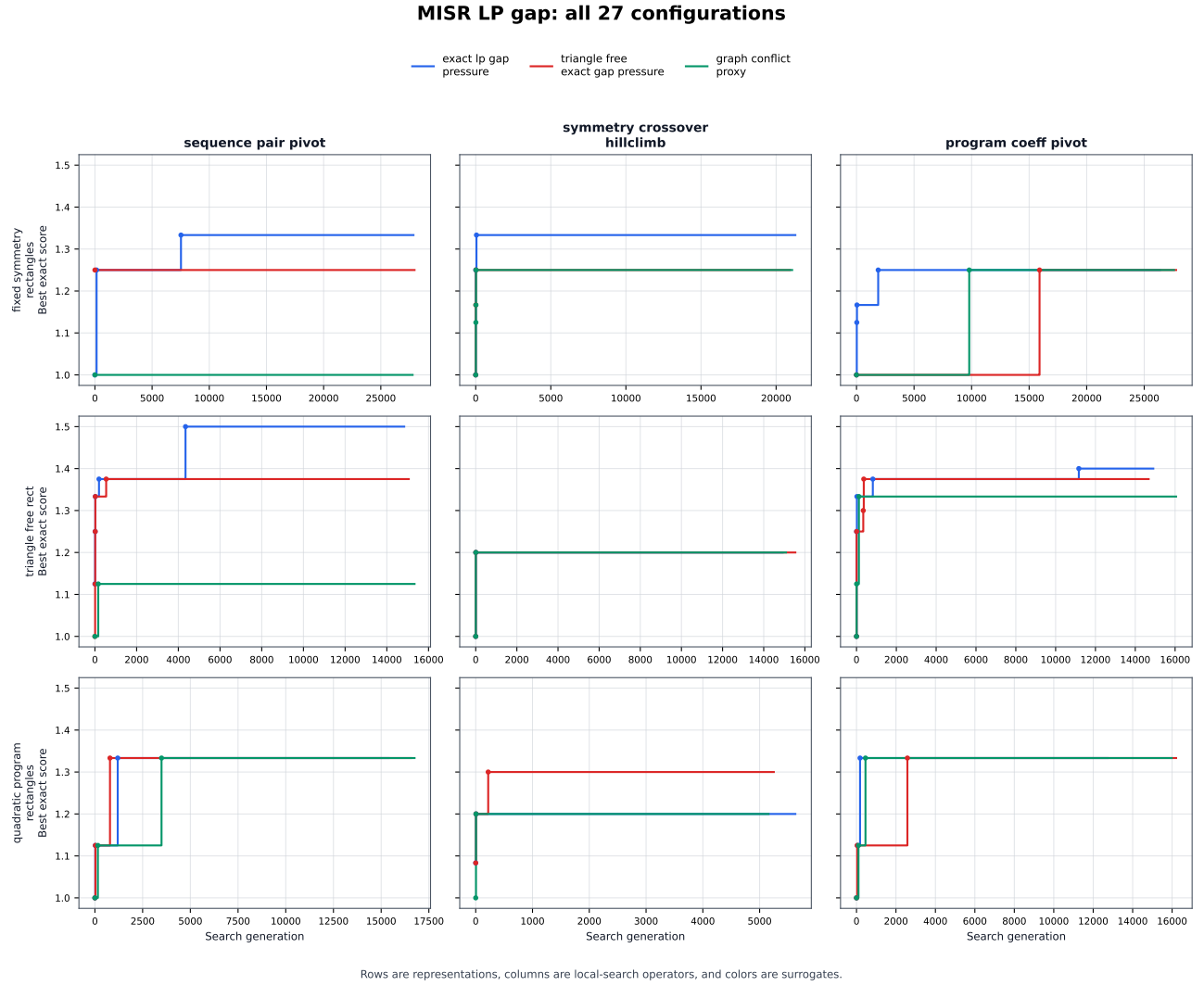


Figure 9: The same 27 MISR cells against search generation. Comparing this figure with Figure 8 distinguishes progress aligned with model updates from raw search-generation progress.

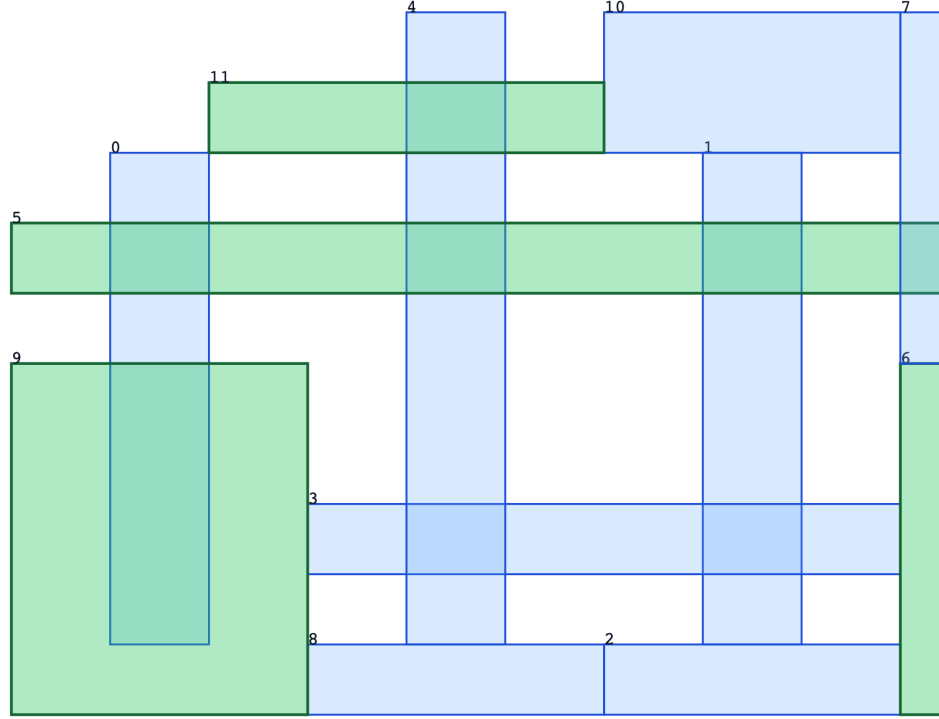


Figure 10: Best audited MISR construction from the final matrix. The drawing is explanatory; the JSON certificate and exact verifier are the evidence.

triangle-free row, `sequence_pair_pivot` upweights add and replace proposals alongside nudge, resize, and delete moves. These moves are useful when they change tight constraints without simultaneously increasing the integral optimum. The $3/2$ record appears at generation 4345, cumulative model epoch 1860, and 3.12 hours.

6.4 Why improvement stopped

Four mechanisms explain the MISR plateau.

1. **Two objectives must move in opposite directions.** Adding intersections can raise LP pressure but often also increases the maximum independent-set structure in a way that leaves the ratio unchanged.
2. **Triangle-free search has a built-in ceiling mechanism.** The clique LP becomes the edge relaxation, and many repaired graphs collapse to common half-integral ratios. Repair can delete exactly the vertices that would have enabled a more unusual denominator.

3. **The representations lack a reliable composition operator.** Duplicating a good finite motif usually scales α and α^* at the same rate, preserving rather than improving the ratio.
4. **Model training is surrogate-conditioned.** If the elite archive contains many geometrically different encodings of the same conflict graph, the transformer learns syntax and local motif frequency more easily than the rare global condition that improves α^*/α .

The result is a discovery method for compact finite witnesses, not yet an amplification mechanism for a new asymptotic lower bound.

Table 8: Descriptive component marginals for Unit-square. Each level pools nine distinct component cells.

Factor	Level	Mean	Median	Min.	Max.	IQR
Representation	<code>sqstab_exact_grid</code>	1.4265	1.4407	1.3158	1.5385	[1.3462, 1.5000]
	<code>line_square_incidence</code>	1.3984	1.4141	1.2500	1.5000	[1.3333, 1.4644]
	<code>fixed_symmetry_grid</code>	1.3960	1.3636	1.2766	1.5000	[1.3043, 1.5000]
Local Search	<code>symmetry_crossover_hillclimb</code>	1.4761	1.5000	1.4141	1.5385	[1.4407, 1.5000]
	<code>sqstab_local_hillclimb</code>	1.3732	1.3462	1.2857	1.5000	[1.3333, 1.3636]
	<code>coord_mutation</code>	1.3715	1.3333	1.2500	1.5000	[1.3043, 1.5000]
Surrogate	<code>exact_stab_gap_pressure</code>	1.4852	1.5000	1.3636	1.5385	[1.5000, 1.5000]
	<code>greedy_partial_lp_bitset</code>	1.3750	1.3636	1.2500	1.5000	[1.3333, 1.4141]
	<code>incidence_statistics</code>	1.3606	1.3333	1.2766	1.5000	[1.3158, 1.4268]

7 Problem 2: unit-square line stabbing

7.1 Objective and exact verifier

For a family $\mathcal{S}$ of congruent axis-parallel squares,

$$\text{score}_{\text{stab}}(\mathcal{S}) = \frac{\tau(\mathcal{S})}{\tau^*(\mathcal{S})},$$

where τ is the minimum number of horizontal and vertical lines stabbing all squares and τ^* is the LP optimum over the compressed critical-line universe. The certificate stores square coordinates, side length, critical lines, line-square incidence, an optimal integer cover, LP primal and dual solutions, and solver status.

7.2 All 27 trajectories

Unit-square stabbing has the largest descriptive marginal contrast in the updated table. The exact gap surrogate averages 1.4852 and every one of its nine cells lies between 15/11 and 20/13. The greedy and incidence-statistics surrogates average 1.3750 and 1.3606. Thus additional generations under a cheap proxy do not compensate for optimizing the wrong quantity. The `sqstab_exact_grid` representation has the strongest representation mean (1.4265), while `symmetry_crossover_hillclimb` has by far the strongest local-search mean (1.4761). Nine cells reach 3/2, and one exact-grid/symmetry-crossover/exact-pressure cell reaches 20/13. Because that winner is also one of the three resolution-island follow-up rows, the record cannot be attributed uniquely to the local-search label.

7.3 What worked

The successful paths optimize the geometric set-cover instance through its critical-line incidence rather than treating coordinates as independent numbers. Exact gap pressure is particularly valuable because

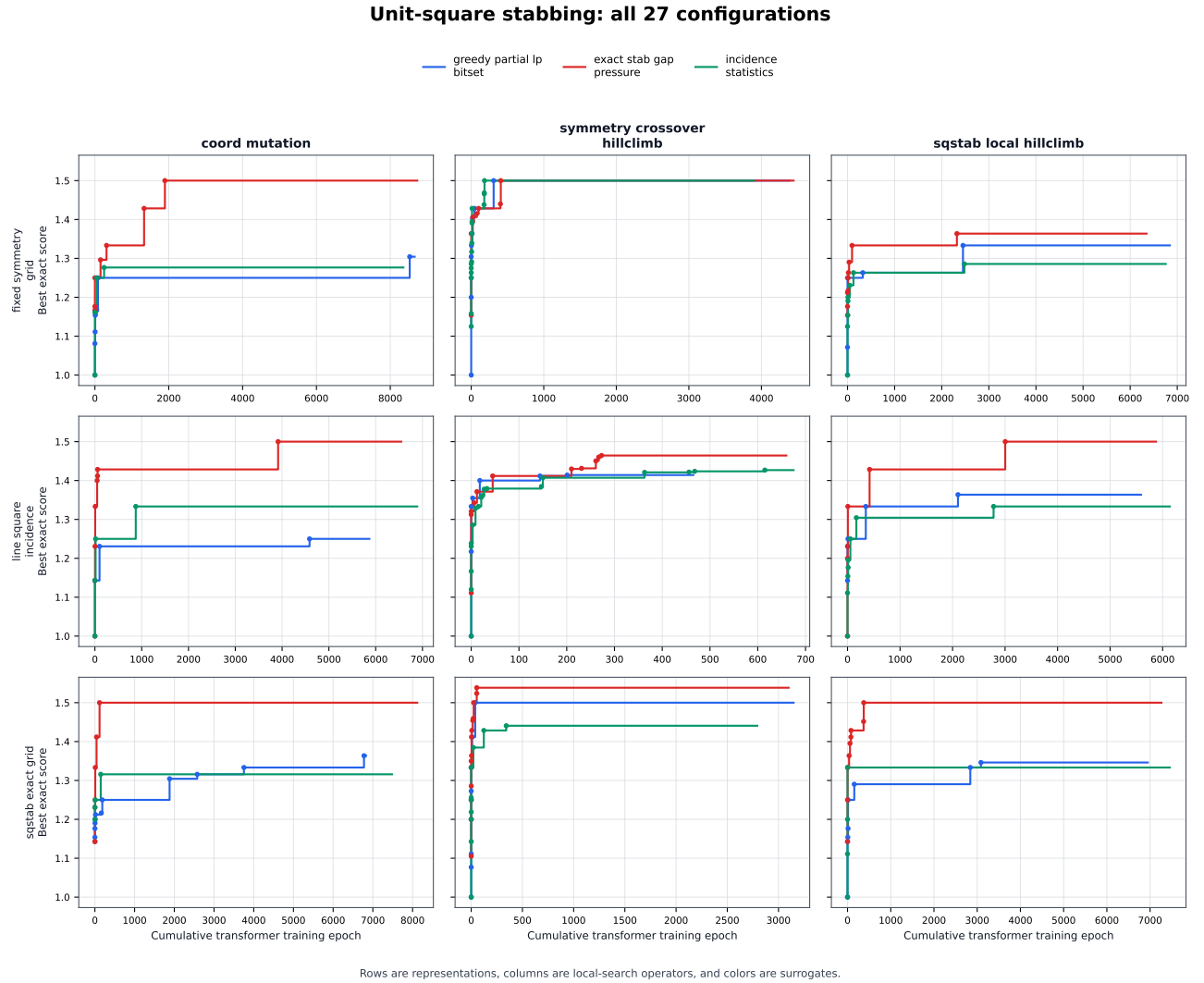


Figure 11: Unit-square trajectories against cumulative transformer-training epoch.

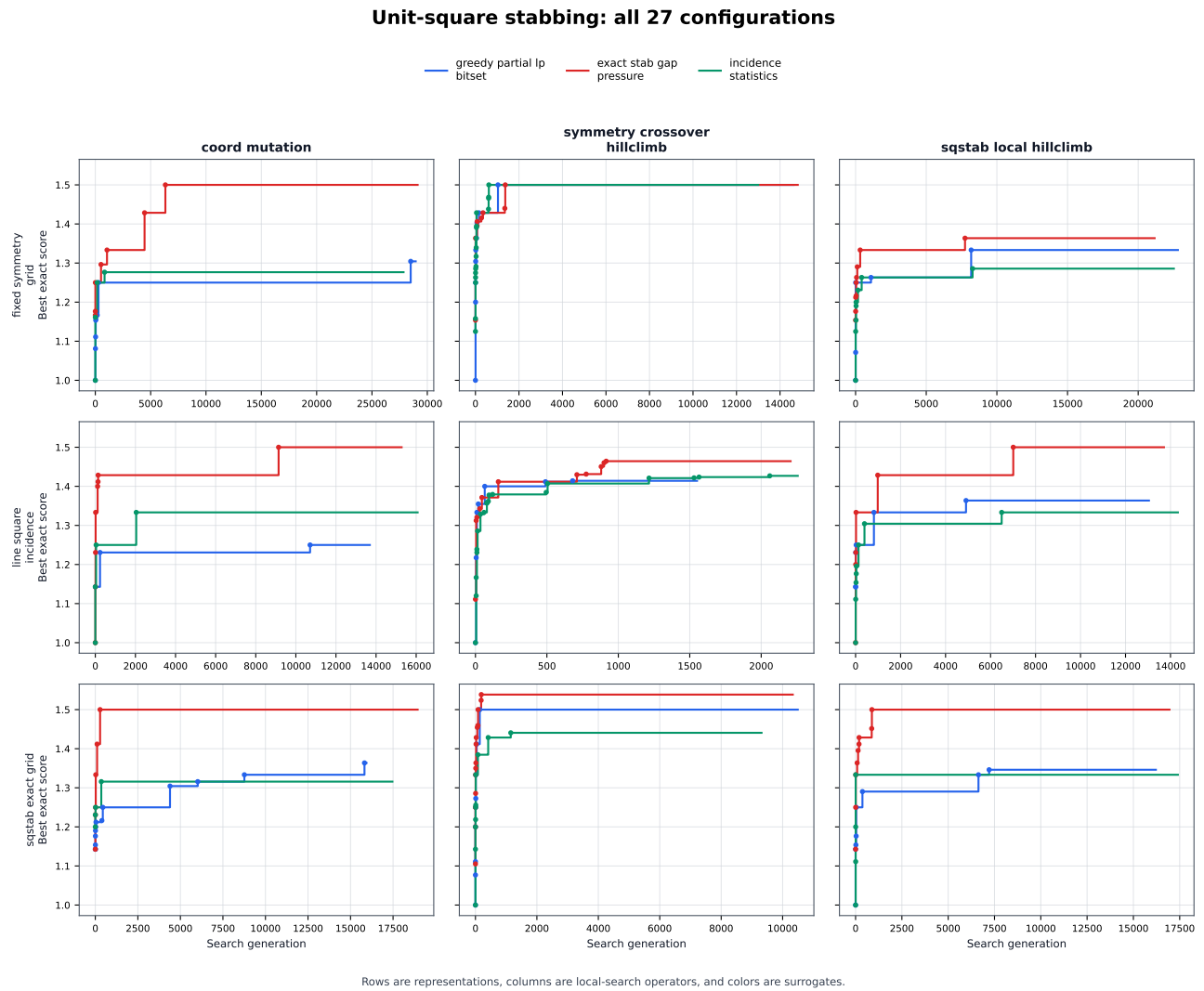


Figure 12: Unit-square trajectories against search generation.

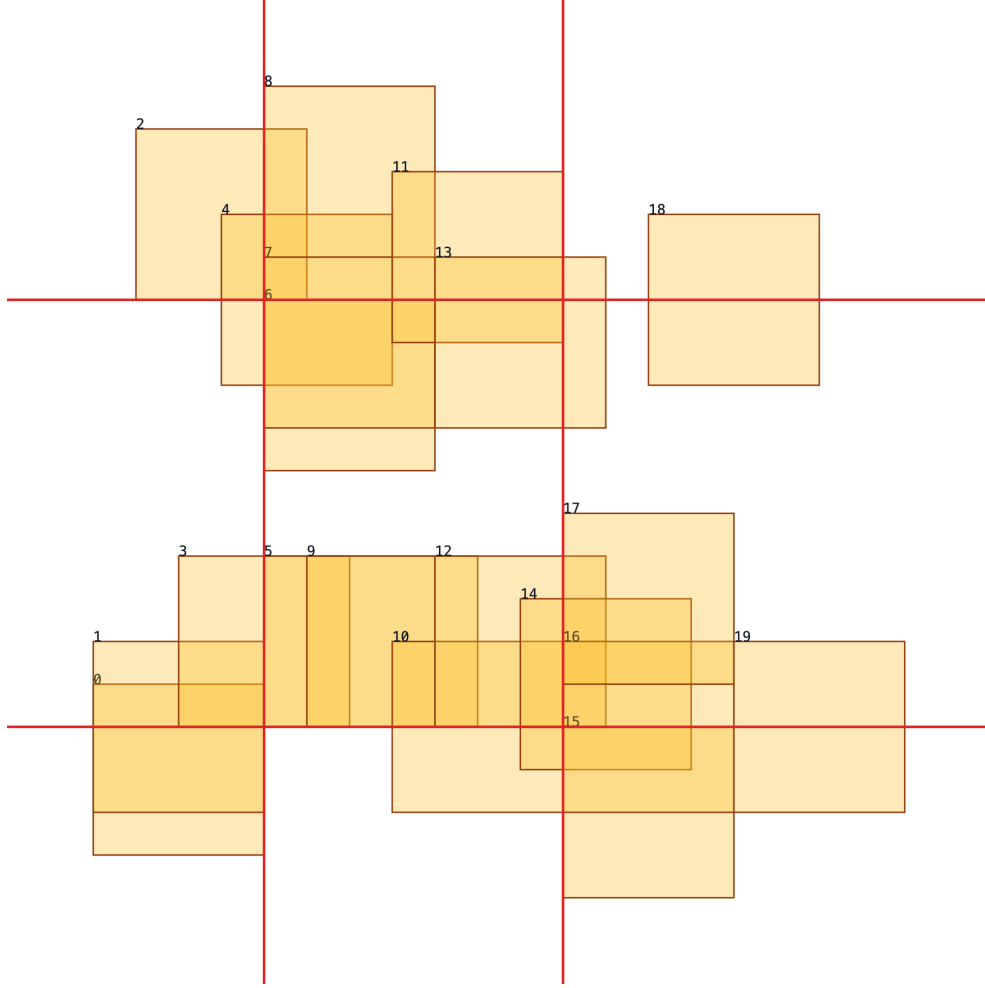


Figure 13: Best audited unit-square stabbing construction from the final matrix. The 20 congruent squares are labeled 0–19; the four red lines form an optimal integral stabbing cover. The exact certificate has $\tau = 4$, $\tau^* = 13/5$, and ratio $20/13$.

greedy cover size, maximum line frequency, and incidence balance are not reliable proxies for the LP optimum. The new hill climb adds coordinated pair exchange and objective-preserving symmetry moves to ordinary shifts and relocations. Its resolution-stratified archive prevents an initially weak side length from being discarded before these moves alter the incidence pattern. The winning $20/13$ certificate has 20 squares, $\tau = 4$, and $\tau^* = 13/5$; it first appears at generation 190, cumulative model epoch 54, after 0.37 hours.

7.4 Why improvement stopped

The unit-square objective is strongly discrete. A local move usually preserves the same critical-line incidence matrix and therefore leaves both τ and τ^* unchanged. A threshold-crossing move often changes the integer and fractional optima together. The final certificates have 9–28 squares on integer coordinate grids, so the search still explores only a small part of the realizable incidence landscape. Moreover, the generic model learns a serialization of coordinates; it has no explicit equivariance under translation, relabeling, or affine normalization.

The $20/13$ result is a rare single-cell hit while $3/2$ is repeatedly rediscovered. The code still has no

neutral, general composition rule that combines such blocks without allowing one line to stab multiple blocks. That cross-block line is the geometric reason direct sums fail to amplify the set-cover gap.

Table 9: Descriptive component marginals for Guillotine. Each level pools nine distinct component cells.

Factor	Level	Mean	Median	Min.	Max.	IQR
Representation	<code>rect_direct_disjoint</code>	0.2349	0.2500	0.1667	0.3333	[0.1667, 0.2727]
	<code>recursive_obstruction_grammar</code>	0.2278	0.2500	0.1667	0.2727	[0.1667, 0.2609]
	<code>fixed_symmetry_packing</code>	0.2222	0.2500	0.1667	0.2500	[0.1667, 0.2500]
Local Search	<code>witness_breaking</code>	0.2674	0.2609	0.2500	0.3333	[0.2500, 0.2727]
	<code>packing_resize</code>	0.2509	0.2500	0.2353	0.2727	[0.2500, 0.2500]
	<code>symmetry_crossover_hillclimb</code>	0.1667	0.1667	0.1667	0.1667	[0.1667, 0.1667]
Surrogate	<code>depth_limited_dp</code>	0.2359	0.2500	0.1667	0.3333	[0.1667, 0.2667]
	<code>first_cut_obstruction</code>	0.2273	0.2500	0.1667	0.2727	[0.1667, 0.2500]
	<code>k_subset_nonseparability</code>	0.2218	0.2500	0.1667	0.2609	[0.1667, 0.2500]

8 Problem 3: recursive guillotine hardness

8.1 Objective and exact verifier

For a pairwise interior-disjoint rectangle family $\mathcal{R}$,

$$\text{score}_{\text{guill}}(\mathcal{R}) = 1 - \frac{\text{save}(\mathcal{R})}{|\mathcal{R}|},$$

where $\text{save}(\mathcal{R})$ is the maximum number of rectangles preserved by an optimal recursive sequence of horizontal and vertical guillotine cuts. The exact dynamic program enumerates coordinate-induced cuts on rectangle-subset states. Its certificate stores the exact saved and destroyed counts, the optimal recursive cut tree, first-cut diagnostics, threshold diagnostics, state count, and solver status.

8.2 All 27 trajectories

The guillotine matrix shows that witness use matters more than a more elaborate generator label. `witness_breaking` averages 0.2674, compared with 0.2509 for `packing_resize`. Every `symmetry_crossover_hillclimb` guillotine cell ends at 1/6, including four structural-random fallback exports, so the shared exact-nonworsening move rule does not transfer successfully to this objective. Among surrogates, `depth_limited_dp` is strongest on average (0.2359), while `k_subset_nonseparability` is weakest (0.2218). Direct disjoint search has the strongest representation mean (0.2349) and contains the unique 1/3 record; the recursive grammar and fixed-symmetry packing average 0.2278 and 0.2222.

8.3 What worked

Guillotine search improves when the representation supplies non-sliceable local structure and the mutation operator responds to actual separable subsets. Witness breaking and packing resize can close

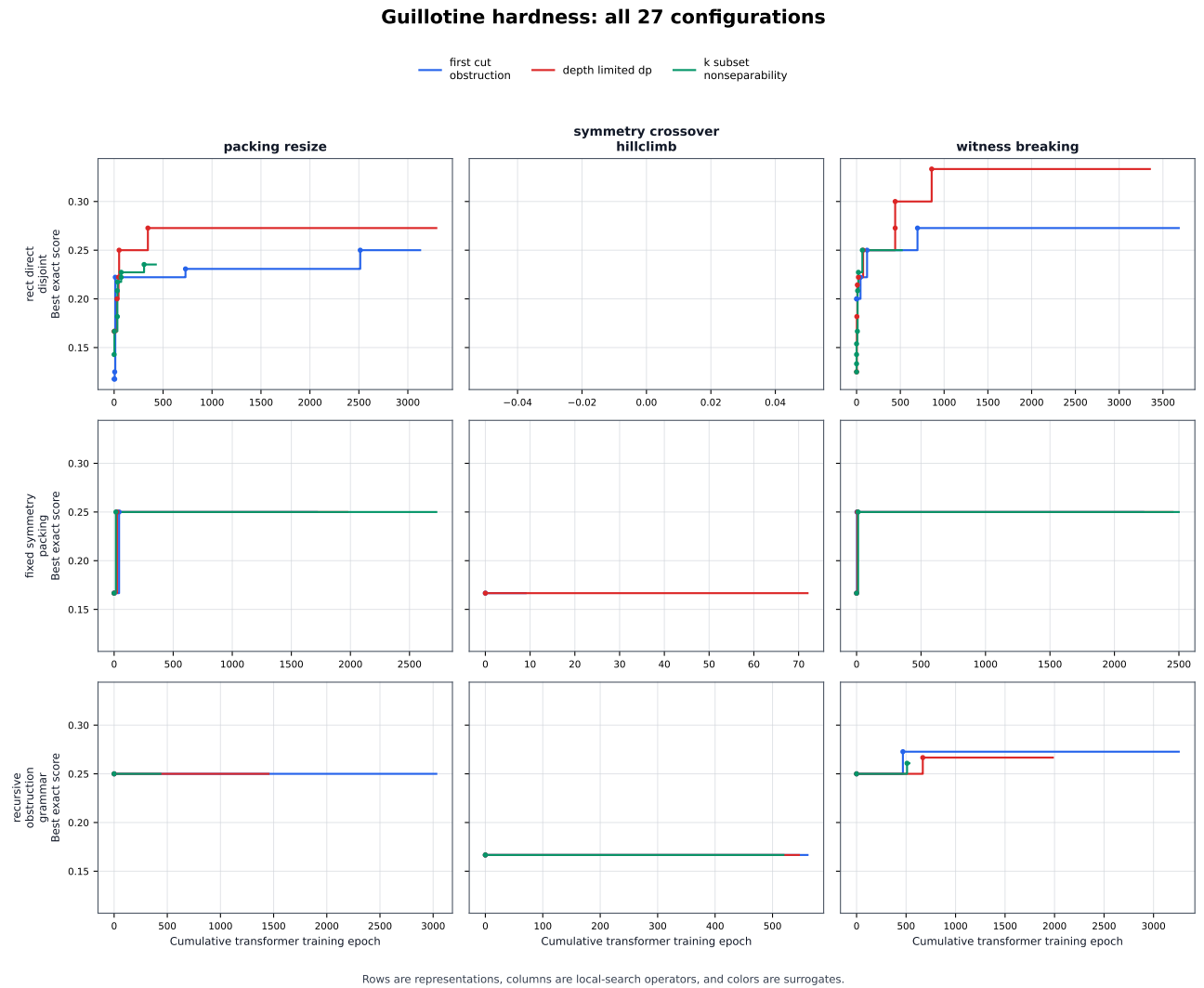


Figure 14: Guillotine trajectories against cumulative transformer-training epoch.

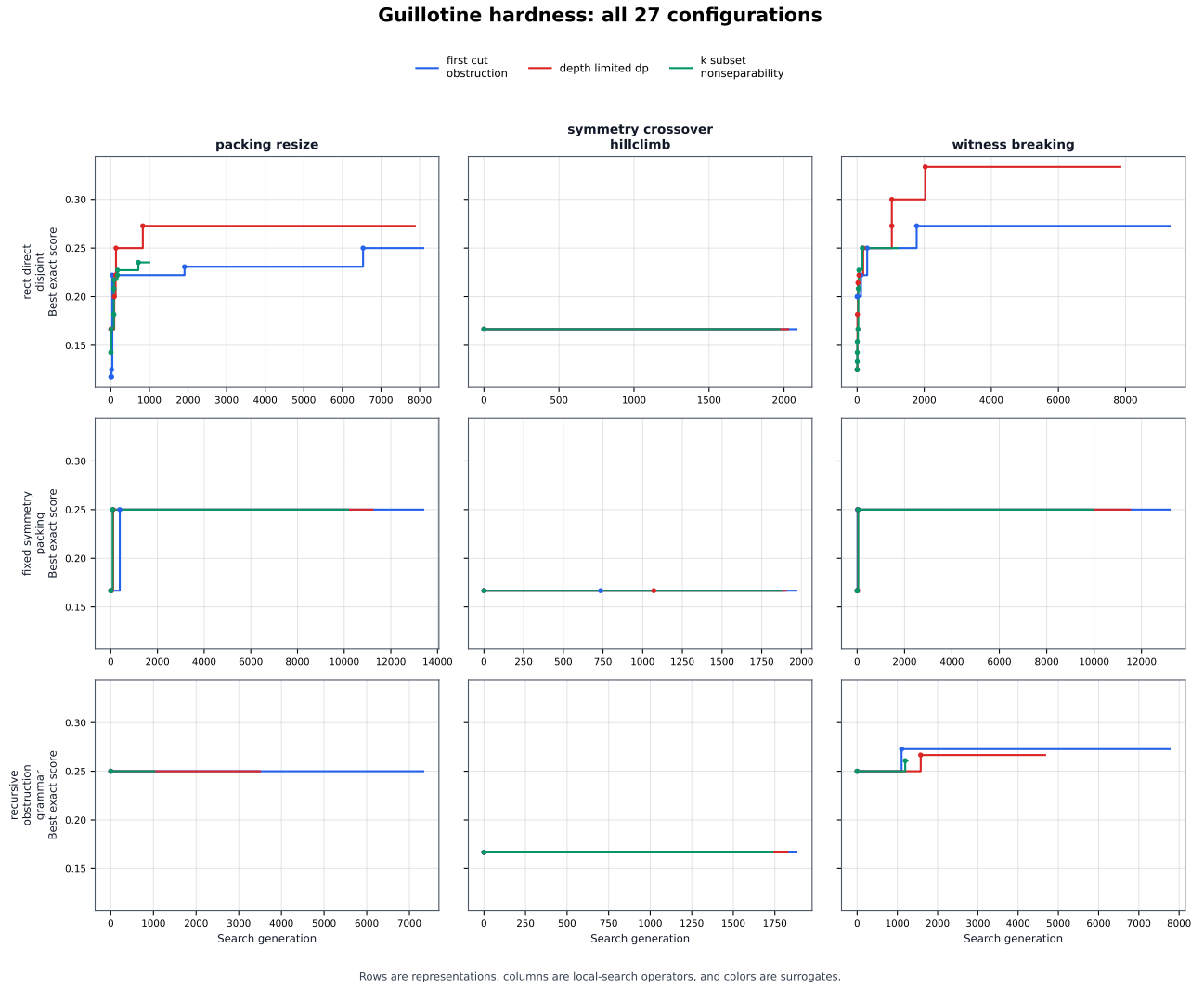


Figure 15: Guillotine trajectories against search generation.

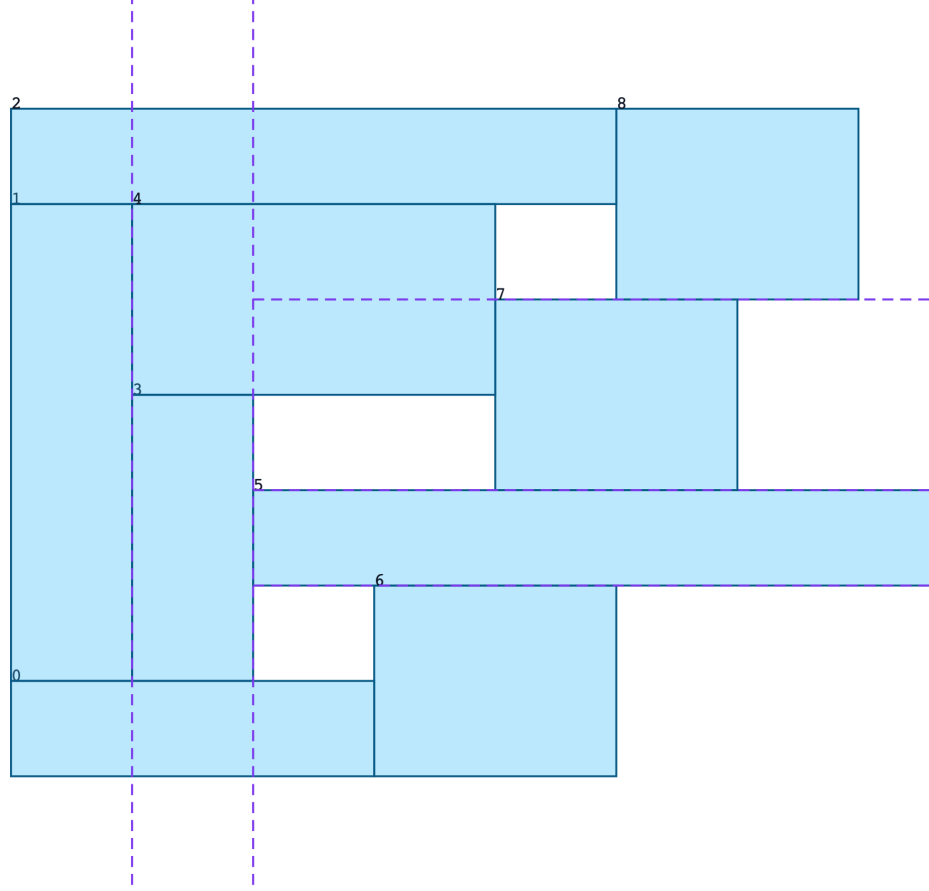


Figure 16: Best audited guillotine construction from the final matrix. The exact cut tree in the certificate, not the drawing, establishes the saved and destroyed counts.

easy projection gaps; the depth-limited dynamic program is more aligned with the exact recursive objective than first-cut obstruction. The k -subset surrogate adds useful large-subset pressure, although it is sampled above 16 rectangles. The winning direct/witness/depth-DP row reaches $1/3$ with nine rectangles, six saved and three destroyed, at generation 2030 and 5.81 hours.

The all-nine $1/6$ result for symmetry/crossover is specific to its replacement runtime and fixed proposal policy. It demonstrates failed transfer in this experiment, not a universal limitation of symmetry moves for guillotine search.

8.4 Why improvement stopped

Guillotine hardness is the clearest case of surrogate mismatch.

1. **First-cut obstruction is not recursive obstruction.** A layout can force an expensive first cut while leaving two easy subinstances.

2. **Depth four is still shallow.** The depth-limited surrogate returns all remaining rectangles as saveable at its horizon, favoring layouts whose weakness appears just beyond that depth.
3. **Large- n threshold tests are sampled.** The deterministic subset schedule is reproducible, but 1024 subsets cannot represent all $\binom{n}{\lfloor n/2 \rfloor + 1}$ possibilities. High sampled nonseparability can coexist with a large exact saved set.
4. **The score has coarse rational steps.** With n rectangles the score changes in increments of $1/n$. Mutations that add easy filler rectangles can make the layout look larger while reducing the destroyed fraction.
5. **Exact feedback arrives only on elites.** The exact cut tree is saved in a certificate but is not fully converted into a differentiable or dense training signal for every candidate.

The fixed spiral topology used by one representation is also a methodological limitation: it can improve the chance of seeing a local obstruction, but it narrows the search toward one family. The representation comparison must therefore be read as “which supplied search family worked best,” not as a neutral estimate over all disjoint rectangle packings.

9 Cross-problem explanation of the barriers

9.1 Prioritized diagnosis

Table 10 ranks the limitations by their likely leverage on future search. The ranking is a mechanistic interpretation of the completed matrix, not a causal effect estimate: there is one stochastic run per cell.

9.2 More epochs cannot repair a closed search space

The transformer samples only instances expressible by the current tokenizer and repaired by the current representation. If no sequence of valid local or model-generated moves reaches a stronger structural family, additional training concentrates probability inside the same basin. A flat learning curve after many successful training events is evidence of this closure. Moreover, the implementation repeatedly trains fresh short-lived models rather than continuing one model, so cumulative epochs do not accumulate weights or optimizer state across events.

9.3 The exact objectives are discontinuous

All three objectives are ratios of optimization values. Small geometric changes often do nothing; the next change may alter both numerator and denominator. Gradient-like local pressure is therefore weak. This also explains why a surrogate can improve smoothly while the exact record remains flat for thousands of generations.

9.4 Repair is both necessary and destructive

Repair enforces triangle freedom, congruent-square validity, disjointness, size bounds, and canonical order. It prevents invalid training data but can erase the coordinated multi-object change proposed by the model. The event logs record repairs and invalid samples; high repair rates indicate that the model is learning a syntax larger than the feasible search manifold.

9.5 The learned representation is mostly syntactic

The transformer is trained on serialized coordinates and permutation-like descriptions rather than on canonical conflict graphs, critical-line incidence structures, LP dual supports, or exact cut trees. Canonicalization removes some trivial equivalences, but many geometrically or combinatorially equivalent candidates still have different strings. This weakens sample efficiency and helps explain why repeated training can produce syntactically valid but structurally redundant candidates.

9.6 Budget enforcement is too coarse

The wall-clock budget is checked between search generations, not inside every surrogate or exact-solver call. A single difficult maximum-independent-set, set-cover, or guillotine-DP evaluation can therefore run beyond the nominal budget and approach the Slurm limit before the row writes its summary. This creates both a robustness risk and a small fairness distortion among cells. Production runs should enforce per-candidate solver deadlines, flush events periodically, and reserve an explicit finalization window.

Table 10: Prioritized limitations supported by the 81-run evidence. “Likely leverage” estimates how strongly fixing the limitation could change the reachable constructions, not how easy the fix would be.

Limitation	Evidence and consequence	Likely leverage
Closed representation families	Entire component families collapse to one score—for example, all nine symmetry/crossover guillotine cells finish at 1/6, and nine MISR cells across several component paths finish at 4/3. Search cannot discover a structure its representation and repair operators cannot express.	Very high
Weak exact-witness feedback	Surrogates rank most candidates, while exact primal, dual, stabbing, and cut-tree witnesses are not converted into dense move-level feedback. Learned-model samples account for only 0, 1, and 1 strict records in MISR, unit-square, and guillotine.	Very high
Archive collapse and model restart	Median duplicate-slot fractions reach 42.6%, 36.1%, and 22.6%. Each training event fits a fresh three-epoch model, so optimizer state and learned features do not accumulate across events.	Very high
No composition mechanism	The best certificates have only 12, 20, and 9 objects. The code searches finite witnesses but has no reliable operator that combines two certified blocks while preserving the relevant ratio. A record therefore does not automatically generate a stronger family.	Very high
Discrete exact objectives	Small geometric changes usually leave the exact ratio unchanged, while the next useful change may require several coordinated objects. This creates sparse rewards and explains the 64–98% median post-best plateaus.	High
Proposal–repair mismatch	Repair guarantees valid triangle-free, unit-square, or disjoint-rectangle instances, but it can undo a coordinated proposal. A high valid-sample rate can therefore coexist with little exact progress.	High
Coordinate serialization rather than invariants	The learned model mainly sees serialized coordinates or permutations. Graph isomorphism, endpoint order, LP-dual support, and cut-tree structure are only partially exposed, so capacity is spent relearning equivalent encodings.	Medium–high
Finite size, grid, and verifier regimes	Object-count and coordinate ranges are bounded to keep exact verification tractable. More epochs at the same scale do not test whether a qualitatively different obstruction appears only at larger n or finer order type.	Medium
Unreplicated cells and runtime confounding	One run per cell prevents error bars over stochastic repetitions. New component levels occur only in the replacement runtime regimes, while solver-dependent throughput gives cells unequal numbers of proposals under the same nominal wall time.	Methodological
Incomplete source provenance	The event label <code>model.sample</code> does not separate a direct transformer sample from the character n -gram backfill. Source-attribution plots are therefore conservative learned-path diagnostics, not a clean transformer ablation.	Measurement

9.7 Archive collapse wastes late training

The training corpus is drawn from the top surrogate-ranked archive without a quality-diversity constraint. Once the archive concentrates around one basin, repeated fresh-model fits generate many duplicates or repaired variants of existing candidates. Duplicate count and valid model-sample yield should therefore be treated as primary learning diagnostics, not merely bookkeeping.

9.8 Finite-scale search may hide a different regime

Exact verification constrains the practical object counts, coordinate grids, and subset limits used during search. Those bounds are scientifically necessary for trustworthy certificates, but they also mean that all 24-hour cells explore a finite regime. A larger obstruction that requires a new order type, denominator, or recursive scale may be absent from every current representation. Progressive size curricula and cross-size transfer should therefore be tested separately from simply extending a fixed-size run.

9.9 One random run per cell limits causal claims

The full factorial permits complete coverage of component combinations, but without repeated seeds an observed difference between two cells contains both a component association and run-specific stochasticity. Marginal means pool nine distinct cells and are useful for prioritization, yet they are not standard errors. This is the principal statistical limitation of the final matrix.

9.10 What better sampling would actually change

Increasing the number of samples from the same archive-conditioned generator would mostly increase duplicates and repaired variants. A useful sampling upgrade would instead combine:

1. a quality-diversity archive indexed by problem-specific structural features rather than surrogate score alone;
2. novelty measured after canonicalization, so translated or relabeled copies do not consume the sampling budget;
3. uncertainty- or disagreement-based exact-audit selection, allocating expensive verifier calls to candidates for which the surrogate is least trusted;
4. adaptive mixtures of model samples, local mutations, and fresh random restarts based on their recent rates of valid and record-improving output; and
5. stratified sampling across object count, denominator, and obstruction type to avoid collapse around one easy rational score.

9.11 What more complex local search should mean

Adding a longer list of arbitrary mutations is unlikely to help. The useful complexity is coordinated and verifier-driven: multi-object moves that attack an LP dual bottleneck, a stabbing line, a separable subset, or a branch in the exact guillotine cut tree. Beam or population search should retain temporary score decreases when they increase structural novelty or close an exact escape witness. This is qualitatively different from applying more single coordinate perturbations.

9.12 The experiment discovers witnesses, not general theorems

An exact certificate proves the objective value of one finite construction. It does not prove an asymptotic family, optimality over all constructions, or a universal lower bound. The missing scientific step for all three problems is composition: a rule that scales a certified finite obstruction while preserving or improving the relevant ratio.

10 Recommendations arising from the data

These recommendations describe what a successor project should change; they are not additional claims from the completed run.

10.1 MISR

1. Learn and mutate canonical conflict graphs jointly with geometric realizations, so graph-equivalent coordinate encodings do not dominate the archive.
2. Use exact LP dual support and the exact independent set to propose coordinated multi-rectangle moves.
3. Test composition operators explicitly by measuring incremental changes in both α^* and α .
4. Separate unrestricted and triangle-free record tracks; their structural barriers differ.

10.2 Unit-square stabbing

1. Search canonical critical-line incidence matrices first, then solve a geometric realization problem.
2. Add incidence-preserving block moves and cross-block separation rules to prevent one line from collapsing a composed construction.
3. Increase coordinate resolution only together with canonicalization; otherwise the model spends capacity on translations and equivalent endpoint orders.
4. Retain exact primal and dual witnesses in the learning features.

10.3 Guillotine

1. Convert the exact optimal cut tree and saved set into explicit mutation targets after every record improvement.
2. Replace fixed-depth scoring with adaptive exact scoring on medium-sized elites and a learned estimator of exact saved fraction for larger candidates.
3. Diversify obstruction generators beyond one fixed topology and report generator family as an explicit experimental factor.
4. Penalize easy filler rectangles and optimize destroyed count and family size lexicographically before taking the ratio.

11 Reproducibility and artifact map

Raw event streams remain in their HPC run roots rather than being committed to Git. After compacting those streams, the final matrix is assembled with the following schematic command; the two angle-bracketed inputs are the compact outputs produced from the corresponding HPC roots:

```
python3 scripts/build_replacement_81_dataset.py \
  --retained-postprocess \
    docs/assets/main_81_24h_20260709_1623/postprocess \
  --delta-postprocess <compact-replacement-42> \
  --delta-postprocess <compact-unit-square-3> \
  --artifact-map \
    docs/assets/replacement_81_final_20260712/run_artifacts/artifact_mapping.csv \
  --out-dir docs/assets/replacement_81_final_20260712/data
```

Plots and descriptive tables are regenerated with:

```
python3 scripts/plot_main_matrix_analysis.py \
  docs/assets/replacement_81_final_20260712/data \
  --out-dir docs/assets/replacement_81_final_20260712/analysis
```

Generated TeX tables are rebuilt from the committed compact evidence with:

```
python3 scripts/generate_main_matrix_tex.py \
  docs/assets/replacement_81_final_20260712/data \
  --plot-table-dir docs/assets/replacement_81_final_20260712/analysis \
  --audit-csv docs/assets/replacement_81_final_20260712/audit/audit.csv \
  --out-dir docs/assets/replacement_81_final_20260712/report
```

The two principal machine-readable artifacts are `docs/assets/replacement\_81\_final\_20260712/data/final\_81\_runs.csv`, with one endpoint row per configuration, and `docs/assets/replacement\_81\_final\_20260712/data/final\_81\_epoch\_history.csv`, with every strict improvement plus an explicit final endpoint. The latter contains generation, elapsed time, cumulative model epoch, source type, and running best score, so score-versus-epoch traces can be reconstructed without the large raw streams. The current plotting script directly consumes `row\_metrics.csv`, `score\_improvements.csv`, and `normalized\_learning\_curves.csv`; the epoch-history file is the portable convenience export. The dataset manifest records row counts, problem coverage, source provenance, and final best values.

A Complete 81-cell table

Table 11: Complete 81-cell result table. Identifiers are shortened here; the full names appear in the design and marginal-effect tables. Plateau is the fraction of completed search generations after the final strict score improvement.

Prob.	Rep.	Local	Surrogate	Best	Gain	Epoch	h	Plat. (%)	n
MISR	fixed_sym_rect	coeff_pivot	exact_lp_gap	1.2500	0.2500	564	0.42	92.9	12
MISR	fixed_sym_rect	coeff_pivot	conflict_proxy	1.2500	0.2500	2937	4.02	64.5	12
MISR	fixed_sym_rect	coeff_pivot	tf_exact_gap	1.2500	0.2500	4770	8.90	42.7	12
MISR	fixed_sym_rect	seq_pivot	exact_lp_gap	1.3333	0.3333	2256	9.12	73.0	12
MISR	fixed_sym_rect	seq_pivot	conflict_proxy	1.0000	0.0000	0	0.00	100.0	12
MISR	fixed_sym_rect	seq_pivot	tf_exact_gap	1.2500	0.0000	0	0.00	100.0	12
MISR	fixed_sym_rect	symmetry_cross	exact_lp_gap	1.3333	0.3333	12	0.30	99.8	12
MISR	fixed_sym_rect	symmetry_cross	conflict_proxy	1.2500	0.2500	3	0.00	99.9	12
MISR	fixed_sym_rect	symmetry_cross	tf_exact_gap	1.2500	0.2500	3	0.00	99.9	12
MISR	qp_rect	coeff_pivot	exact_lp_gap	1.3333	0.3333	75	0.29	98.6	8
MISR	qp_rect	coeff_pivot	conflict_proxy	1.3333	0.3333	192	0.33	97.2	8
MISR	qp_rect	coeff_pivot	tf_exact_gap	1.3333	0.3333	1104	2.11	84.1	8
MISR	qp_rect	seq_pivot	exact_lp_gap	1.3333	0.3333	510	1.19	92.2	8
MISR	qp_rect	seq_pivot	conflict_proxy	1.3333	0.3333	1494	3.02	79.2	8
MISR	qp_rect	seq_pivot	tf_exact_gap	1.3333	0.3333	336	0.60	95.2	8
MISR	qp_rect	symmetry_cross	exact_lp_gap	1.2000	0.1167	0	0.02	99.9	12
MISR	qp_rect	symmetry_cross	conflict_proxy	1.2000	0.2000	0	0.02	99.9	12
MISR	qp_rect	symmetry_cross	tf_exact_gap	1.3000	0.2167	63	0.91	95.8	12
MISR	triangle_free	coeff_pivot	exact_lp_gap	1.4000	0.4000	4782	14.48	25.1	14
MISR	triangle_free	coeff_pivot	conflict_proxy	1.3333	0.3333	51	0.05	99.3	8
MISR	triangle_free	coeff_pivot	tf_exact_gap	1.3750	0.3750	156	0.15	97.5	11
MISR	triangle_free	seq_pivot	exact_lp_gap	1.5000	0.3750	1860	3.12	70.7	12
MISR	triangle_free	seq_pivot	conflict_proxy	1.1250	0.1250	66	0.06	99.0	9
MISR	triangle_free	seq_pivot	tf_exact_gap	1.3750	0.3750	231	0.22	96.4	11
MISR	triangle_free	symmetry_cross	exact_lp_gap	1.2000	0.2000	0	0.00	100.0	12
MISR	triangle_free	symmetry_cross	conflict_proxy	1.2000	0.2000	0	0.00	100.0	12
MISR	triangle_free	symmetry_cross	tf_exact_gap	1.2000	0.2000	0	0.01	100.0	12
Unit sq.	fixed_sym_grid	coord_mut	exact_stab_gap	1.5000	0.3333	1896	8.36	78.3	20
Unit sq.	fixed_sym_grid	coord_mut	greedy_lp	1.3043	0.3043	8526	23.32	1.6	20

Prob.	Rep.	Local	Surrogate	Best	Gain	Epoch	h	Plat. (%)	n
Unit sq.	fixed_sym_grid	coord_mut	incidence_stats	1.2766	0.2766	249	0.18	97.0	20
Unit sq.	fixed_sym_grid	sqstab_hill	exact_stab_gap	1.3636	0.2098	2322	5.56	63.4	20
Unit sq.	fixed_sym_grid	sqstab_hill	greedy_lp	1.3333	0.3333	2451	4.87	64.1	20
Unit sq.	fixed_sym_grid	sqstab_hill	incidence_stats	1.2857	0.2857	2481	4.88	63.2	20
Unit sq.	fixed_sym_grid	symmetry_cross	exact_stab_gap	1.5000	0.3462	408	2.06	90.8	20
Unit sq.	fixed_sym_grid	symmetry_cross	greedy_lp	1.5000	0.5000	309	2.66	92.9	20
Unit sq.	fixed_sym_grid	symmetry_cross	incidence_stats	1.5000	0.3750	183	0.92	95.3	20
Unit sq.	line_sq_inc	coord_mut	exact_stab_gap	1.5000	0.5000	3915	11.18	40.2	21
Unit sq.	line_sq_inc	coord_mut	greedy_lp	1.2500	0.1071	4587	16.21	21.8	24
Unit sq.	line_sq_inc	coord_mut	incidence_stats	1.3333	0.3333	873	1.01	87.3	9
Unit sq.	line_sq_inc	sqstab_hill	exact_stab_gap	1.5000	0.5000	3003	9.29	48.9	21
Unit sq.	line_sq_inc	sqstab_hill	greedy_lp	1.3636	0.2208	2103	6.05	62.4	24
Unit sq.	line_sq_inc	sqstab_hill	incidence_stats	1.3333	0.3333	2781	7.03	54.7	15
Unit sq.	line_sq_inc	symmetry_cross	exact_stab_gap	1.4644	0.3532	273	11.67	58.5	28
Unit sq.	line_sq_inc	symmetry_cross	greedy_lp	1.4141	0.4141	201	1.52	56.2	20
Unit sq.	line_sq_inc	symmetry_cross	incidence_stats	1.4268	0.4268	615	20.80	8.7	20
Unit sq.	sqstab_grid	coord_mut	exact_stab_gap	1.5000	0.3571	117	0.35	98.5	14
Unit sq.	sqstab_grid	coord_mut	greedy_lp	1.3636	0.2208	6777	23.68	0.8	20
Unit sq.	sqstab_grid	coord_mut	incidence_stats	1.3158	0.1158	147	0.14	98.0	10
Unit sq.	sqstab_grid	sqstab_hill	exact_stab_gap	1.5000	0.3571	375	0.46	94.8	19
Unit sq.	sqstab_grid	sqstab_hill	greedy_lp	1.3462	0.3462	3087	6.68	55.5	22
Unit sq.	sqstab_grid	sqstab_hill	incidence_stats	1.3333	0.3333	0	0.00	100.0	9
Unit sq.	sqstab_grid	symmetry_cross	exact_stab_gap	1.5385	0.4332	54	0.37	98.2	20
Unit sq.	sqstab_grid	symmetry_cross	greedy_lp	1.5000	0.5000	39	1.41	98.7	20
Unit sq.	sqstab_grid	symmetry_cross	incidence_stats	1.4407	0.4407	342	2.39	87.7	20
Guill.	fixed_sym_pack	pack_resize	depth_dp	0.2500	0.0833	24	0.17	99.1	12
Guill.	fixed_sym_pack	pack_resize	first_cut	0.2500	0.0833	42	0.42	97.1	12
Guill.	fixed_sym_pack	pack_resize	k_subset	0.2500	0.0833	15	0.17	99.2	12
Guill.	fixed_sym_pack	symmetry_cross	depth_dp	0.1667	0.0000	0	13.24	43.7	12
Guill.	fixed_sym_pack	symmetry_cross	first_cut	0.1667	0.0000	0	8.90	62.7	12
Guill.	fixed_sym_pack	symmetry_cross	k_subset	0.1667	0.0000	0	24.01	0.0	12
Guill.	fixed_sym_pack	witness_break	depth_dp	0.2500	0.0833	6	0.06	99.7	12

Prob.	Rep.	Local	Surrogate	Best	Gain	Epoch	h	Plat. (%)	n
Guill.	fixed_sym_	witness_	first_cut	0.2500	0.0833	3	0.02	99.9	12
	pack	break							
Guill.	fixed_sym_	witness_	k_subset	0.2500	0.0833	12	0.09	99.5	12
	pack	break							
Guill.	rect_direct	pack_resize	depth_dp	0.2727	0.1061	345	1.92	89.5	11
Guill.	rect_direct	pack_resize	first_cut	0.2500	0.1324	2514	17.91	19.3	8
Guill.	rect_direct	pack_resize	k_subset	0.2353	0.0924	306	16.72	28.3	17
Guill.	rect_direct	symmetry_	depth_dp	0.1667	0.0000	0	24.01	0.0	12
		cross							
Guill.	rect_direct	symmetry_	first_cut	0.1667	0.0000	0	24.01	0.0	12
		cross							
Guill.	rect_direct	symmetry_	k_subset	0.1667	0.0000	0	24.01	0.0	12
		cross							
Guill.	rect_direct	witness_	depth_dp	0.3333	0.2083	858	5.81	74.1	9
		break							
Guill.	rect_direct	witness_	first_cut	0.2727	0.0727	696	3.06	80.9	11
		break							
Guill.	rect_direct	witness_	k_subset	0.2500	0.1250	66	2.74	87.2	8
		break							
Guill.	obstruction_	pack_resize	depth_dp	0.2500	0.0000	0	0.00	100.0	8
	grammar								
Guill.	obstruction_	pack_resize	first_cut	0.2500	0.0000	0	0.00	100.0	8
	grammar								
Guill.	obstruction_	pack_resize	k_subset	0.2500	0.0000	0	0.00	100.0	8
	grammar								
Guill.	obstruction_	symmetry_	depth_dp	0.1667	0.0000	0	0.00	100.0	12
	grammar	cross							
Guill.	obstruction_	symmetry_	first_cut	0.1667	0.0000	0	0.00	100.0	12
	grammar	cross							
Guill.	obstruction_	symmetry_	k_subset	0.1667	0.0000	0	0.00	100.0	12
	grammar	cross							
Guill.	obstruction_	witness_	depth_dp	0.2667	0.0167	669	8.10	66.2	15
	grammar	break							
Guill.	obstruction_	witness_	first_cut	0.2727	0.0227	468	2.54	85.8	11
	grammar	break							
Guill.	obstruction_	witness_	k_subset	0.2609	0.0109	510	22.87	4.6	23
	grammar	break							

B Interpretation guide

Table 12: How to interpret the main analysis fields.

Field	Meaning
Best exact score	Highest score recomputed by the problem’s exact scorer and saved in the row checkpoint/summary.
Initial best	First nontrivial exact score visible in the event stream.
Score gain	Final best minus initial best; it measures within-run discovery, not comparison with historical experiments.
Best model epoch	Cumulative successful transformer-training epochs before the event that first exposed the final best.
Time to best	Wall-clock seconds from row start to the first saved final-best certificate.
Plateau fraction	Fraction of completed search generations occurring after the last strict best-score improvement.
Model improvement	Strict improvement whose audited candidate had <code>source_type=model_sample</code> .
Fallback	Fresh structural-random export attempted only when a guillotine row would otherwise end without any certificate; used by four poor-performing symmetry/crossover rows and never by a reported record.

C Evidence boundary

The report excludes epsilon-net and graph-separation experiments, controls from earlier arrays, all square-stabbing-14-9 package evidence, historical scores without certificates, and warm-start runs. Those artifacts may be discussed elsewhere, but they are not mixed into this 81-cell final matrix.